

Consumption tax and corporate product mix decisions[☆]Pulak Ghosh^a, Martin Jacob^{b,*}, Ya Kang^c, Jian Zhang^d^a Indian Institute of Management Bangalore, India^b IESE Business School, Spain^c Chinese University of Hong Kong, Hong Kong^d Faculty of Business and Economics, The University of Hong Kong, Pokfulam Road, Hong Kong Special Administrative Region of China

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ABSTRACT

This paper examines how consumption taxes—through the taxation of input goods—can affect firms' product mix decisions. We study India's staggered transition from a sales tax, where there was the risk that intermediate inputs are taxed multiple times along the supply chain, to a value-added tax (VAT) that offered credits for taxes paid on inputs. Using detailed product-level data for manufacturing firms and a stacked difference-in-differences design, we find that firms respond to the VAT adoption by narrowing their product scope through reducing vertical integration. These changes reflect a shift away from internal production of inputs, which was tax-induced under the sales tax system, toward greater specialization in the most profitable outputs. We also document that this vertical disintegration leads to lower manufacturing costs, greater profitability, improved investment efficiency, and higher firm value. Our findings highlight how consumption tax design can affect resource allocation within firms and that reducing tax burdens on input goods enhances operational efficiency.

1. Introduction

Consumption taxes represent the largest source of tax revenue for governments globally. In OECD countries, they account for roughly one-third of total tax revenues and generate, on average, 9.2% of GDP—more than twice the revenue from corporate income taxes (OECD, 2024).¹ Importantly, because consumption taxes are levied on firms' outputs, they have the potential to fundamentally reshape business behavior by directly influencing cost structures, profitability, and strategic decision-making. Despite their importance, relatively little is known about how consumption taxes shape corporate behavior and resource allocation within and across firms. While evidence exists on the investment effects of consumption taxes (e.g., Jacob et al., 2019; Jacob et al., 2024), the accounting literature on the real effects of non-income taxes remains at an early stage (e.g., Jacob, 2022; Dyreng et al., 2025; Lester and Olbert, 2025). We advance this emerging literature by exploring the effect of consumption taxes on firms' product portfolio choices, which lie

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¹ Consumption taxes also serve as a fiscal policy tool for welfare redistribution. When properly designed, they can be as redistributive as income taxes in reducing inequality and increasing welfare (Kato, 2003; Bachas et al., 2024).

at the core of business strategy. Product portfolio decisions shape how firms allocate resources, manage risk, and make investments. Understanding how taxes influence these choices is thus crucial for academics and policymakers, in particular, because input tax credits under a value-added tax (VAT) can alter the relative cost of internal versus external production and thus affect the optimal degree of vertical integration.

In this paper, we examine how consumption taxes—by taxing input goods—can change a firm's product mix by affecting its decision to vertically integrate. To illustrate, consider a firm that uses a single input good to produce a final good sold to consumers at a price of 200. The input good is either sourced externally at a price of 100 or produced internally at a cost of 100. The firm is therefore indifferent between vertical integration (i.e., producing both the input and the final good) and external sourcing, as both strategies yield a profit of 100. In theory, neither a sales tax nor a VAT distorts this decision. With a tax rate of 10%, the final consumer pays 220, and the firm earns a profit of 100 under both a sales tax system and a VAT system.² In practice, however, implementing a sales tax without taxing input goods is challenging because distinguishing taxable consumers from tax-exempt businesses is not straightforward.³ As a result, sales taxes may be paid not only by consumers but also by businesses on input goods, leading to potentially multiple layers of taxation, or “tax cascading.” This additional tax burden on externally sourced input goods affects a firm's product mix decision. To avoid this burden—and to maintain a final consumer price of 220 and a profit of 100 in our example—the firm has a tax incentive to vertically integrate its supply chain and produce input goods internally. In contrast, a VAT with full and immediate input tax credits creates no incentive for vertical integration because it imposes no additional tax burden on externally sourced goods relative to internally produced goods.⁴

Despite the economic significance of consumption taxes and their potential to shape firms' product mix decisions when input goods are taxed, empirically testing this notion faces at least two challenges. First, measuring product portfolios and related outcomes requires granular firm–product-level data, including information on internally produced input goods (Banker et al., 2018). Such detailed data, however, are rarely available. Second, credible empirical testing requires (i) meaningful variation in the tax burden on input goods and (ii) a credible identification strategy to isolate the effect of this variation on product mix decisions.

We overcome these challenges by leveraging a unique empirical setting in India that allows us to examine how tax burdens on input goods affect firms' product mix decisions. First, we use unique panel data that track all products—including input goods—produced by all listed Indian manufacturing firms. These data include product names, types, manufacturing costs, quantities, and selling prices. This level of granularity, combined with mandatory reporting requirements, allows us to observe firms' product portfolio decisions with minimal measurement error. Second, we exploit one of the largest consumption tax reforms in Indian history: the staggered introduction of the VAT, which replaced the former sales tax system that generated cascading taxes by taxing input goods. The new VAT regime—while not entirely frictionless—substantially reduced the tax burden on input goods by granting firms input tax credits.⁵ Because VAT adoption occurred at different times across Indian states, we identify the effect of VAT adoption—that is, the reduction in the tax burden on input goods—on firms' product mix decisions using a stacked difference-in-differences (DiD) approach (e.g., Cengiz et al., 2019; Baker et al., 2022). Importantly, prior work shows that cross-state variation in the timing of VAT adoption was driven by political rather than economic factors (Aneja et al., 2021). This combination of rich data and a large policy reform makes India an ideal setting to study how consumption tax burdens on input goods affect firms' product portfolio choices through decisions to vertically integrate.

However, we note that focusing on a single developing country may limit the generalizability of our findings, particularly to developed countries such as the U.S. Nevertheless, India's role as the world's fifth-largest manufacturer and as a major emerging market makes our findings potentially relevant to other developing countries with similar institutional frictions. Importantly, the mechanism we uncover—distortions arising from levying a consumption burden on input goods—is a common issue in the consumption tax systems of many developing countries, where delayed or incomplete input tax credits often create tax burdens on input goods and lead to tax cascading (see, e.g., IMF, 2021; Brockmeyer et al., 2024).

We begin by showing how the switch from sales tax to VAT affects firms' product mix. Our main outcome variable captures the number of unique product types. For example, a car manufacturer producing multiple models of cars would be counted as offering only one product type. If the car manufacturer also produces input goods (e.g., car doors, belts, or brakes), our product mix measure indicates that the firm produces multiple unique product types. Our findings show that the transition to VAT leads to a decline in the number of product types. Economically, we estimate a semi-elasticity of -0.8 to -1 for the number of products with respect to the

² Under a sales tax, the input good is tax exempt, and only the final pretax selling price of 200 is subject to tax. The consumer therefore pays 220 in total, of which 20 is sales tax. Under a VAT, the tax on external inputs is 10. The firm pays 110 to the supplier but receives an input tax credit of 10, resulting in an after-tax input cost of 100. When the firm sells the final good to the consumer at a pretax price of 200, the consumer again pays 220, including 20 in VAT.

³ Consistent with the notion that firms also bear sales taxes, research on the U.S. and Canada finds that more than 40% of sales tax revenues are paid by firms (Ring Jr, 1999; Bird and Smart, 2014). Similarly, in Malaysia, which operates a sales tax system, practitioners identify tax cascading as a key issue (see, for example, <https://www.pwc.com/my/en/assets/media/pwc-in-the-news/2023/pwc-my-061123-the-invisible-cost-of-sst.pdf>).

⁴ Importantly, a VAT can also impose a tax burden on externally sourced inputs when input tax credits are not fully granted or are refunded with substantial delays. The International Monetary Fund notes severe problems related to delayed refunds and the nonpayment of excess input VAT credits despite legitimate claims (IMF, 2021). While these issues are less prevalent in high-income countries, they remain substantial in upper-middle- and lower-income countries.

⁵ Despite being a first-point-of-sale setup in India, the sales tax was essentially levied at multiple production stages (Agrawal and Zimmermann, 2025). According to India's official tax policy description, the VAT White Paper (2005), one of the major motivations for switching to VAT was to eliminate the cascading effect of sales taxes levied at every point along the production chain.

cascading sales tax. This semi-elasticity is sizable and larger than recent estimates of the semi-elasticity of investment (Chodorow-Reich, 2025). At the same time, we note that our semi-elasticity is smaller than that for export revenue recognition and employee relocation within Europe in De Simone and Olbert, 2026, for profit shifting in Delis et al. (2025), or for foreign direct investment (e.g., Feld and Heckemeyer, 2011). The effect is not explained by time-varying industry factors, time-invariant firm characteristics, or contemporaneous state-level economic policies that could be associated with firms' product mix strategies and that may coincide with VAT changes.⁶ We also show that treated and control firms follow parallel pre-trends prior to VAT adoption. After VAT adoption, treated firms reduce their product offerings. In addition, following Berg et al. (2021), we show that our estimates are unlikely to be biased by potential spillover effects on the control group.

Next, to shed light on the mechanism through which the VAT adoption affects product mix decisions, we provide more direct evidence of vertical disintegration. We construct a firm-level vertical integration measure by leveraging a large language model (LLM) and the detailed product descriptions in our product-level data. The resulting measure captures the extent to which a firm's products have input–output relationships with one another. Consistent with our main argument, we find that firms respond to VAT adoption by reducing their degree of vertical integration. Moreover, the effect of VAT adoption on product mix is primarily driven by firms that are *ex ante* vertically integrated. We document a larger economic magnitude of the effect for vertically integrated firms than for the full sample. In contrast, horizontally integrated firms do not change their product mix.

In the next step, we unpack the main effect by examining firms' adjustments in product mix *within* product types. While the main tests show that firms reduce the number of broad product types offered (e.g., cars, tires, brakes, and doors for car manufacturers), VAT adoption may incentivize firms to specialize in the most profitable products within each retained product type. For example, a car manufacturer might expand its range of car models (e.g., adding a sports car to a sedan lineup) while dropping less profitable product types, such as tires. Using detailed data on individual products and their markups, we find that firms introduce more subtypes of the most profitable products within the product types they continue to produce. In other words, treated firms reduce the number of product types while expanding the variety of subtypes within the most profitable retained product types.

We also conduct several cross-sectional analyses to corroborate our main findings and explore their theoretical underpinnings. First, we examine the role of tax incidence (e.g., Weyl and Fabinger, 2013). The impact of VAT adoption depends on the extent to which the tax incidence falls on the focal firm versus its suppliers or customers. If the sales tax burden is borne more by the firm than by its customers or suppliers, we expect a stronger effect of VAT adoption on product mix. To test this, we use an industry-level proxy for tax incidence (market concentration) and various firm-level proxies to capture differences in the tax incidence borne by firms. We find that the effect of VAT adoption on product mix is (i) more pronounced for firms operating in more concentrated markets and (ii) attenuated when consumers or suppliers bear more of the tax burden. Second, we exploit differences in labor intensity. Firms with labor-intensive products are more likely to outsource production after VAT adoption because they become eligible for input tax credits on external labor costs. Consistent with this notion, we find a stronger effect of VAT adoption on product mix for firms with more labor-intensive products.

Finally, we examine the effect of VAT adoption on profitability and investment efficiency. Under a sales tax, firms maximize value by vertically integrating, i.e., by producing input goods to avoid cascading taxes. Because the extent of vertical integration is partly tax-induced, it does not reflect the optimal allocation of resources. The transition to VAT allows firms to adopt a level of vertical integration that is closer to the optimal product mix from a nontax perspective. Consistent with this notion, we find lower manufacturing costs following VAT adoption, suggesting that the cascading sales tax inefficiently pushed firms toward costlier internal production of input goods. We also find higher profitability and firm value, consistent with less distorted production decisions after VAT adoption. To further corroborate this interpretation, we explore the impact on investment efficiency. Under a cascading sales tax, firms may be constrained in responding to investment opportunities because they may need to invest in multiple vertically integrated products when new opportunities arise. In contrast, a VAT system allows firms to respond more flexibly to investment opportunities. Using investment-Q sensitivity as our measure of investment efficiency (e.g., Lucas Jr and Prescott, 1971; Whited, 1992; Edmans et al., 2017; Jayaraman and Wu, 2019), we show that firms exhibit higher investment-Q sensitivity after the reform, indicating higher investment efficiency under the less distortive VAT regime.

With these findings, our study expands previous research on firms' responses to taxation, including the effects of corporate taxes (e.g., Djankov et al., 2010; Ljungqvist et al., 2017; Langenmayr and Lester, 2018; Giroud and Rauh, 2019; Lester, 2019; Maffini et al., 2019; Kang et al., 2021; Jacob, 2022; Gallemore et al., 2025; Lester and Olbert, 2025), dividend taxes (Becker et al., 2013; Yagan, 2015), and tax credits (Rao, 2016). The literature on corporate taxes has mostly focused on the impact of taxes on firms' investments, using various identification strategies and administrative tax data (e.g., Zwick and Mahon, 2017; Liu and Mao, 2019; Chen et al., 2021). Moreover, despite the importance of non-income taxes for government revenue, only a few papers examine non-income taxes (Dyreng et al., 2025). These papers focus primarily on tax planning with non-income taxes in international settings (Robinson, 2012; De Simone and Olbert, 2026) or in the U.S. (Fox et al., 2022; Fox et al., 2023). Our main contribution is that we explore a new dimension of the real effects of tax policy. We show that firms' product portfolios are affected by consumption taxes and that reducing tax burdens on input goods mitigates the adverse effects of consumption taxes on product decisions and can increase economic efficiency.

In this regard, our study responds to the call in Lester and Olbert (2025) for more rigor in measuring investment and real outcomes and to the call in Dyreng et al. (2025) to explore non-income taxes. We use detailed product-level data to examine the real economic

⁶ We also implement a specification with state–year fixed effects to account for both observable and unobservable state-year confounding factors in the cross-sectional tests. Our main inference remains robust.

responses (i.e., the product mix of manufacturing firms) to changes in consumption tax regimes. Our use of administrative data advances prior studies that need to rely on firms' voluntary disclosures to measure product portfolios. These data allow us to capture the actual production scope and the extent of vertical integration decisions. This approach provides a more accurate picture of how firms adjust their operations and differentiate their real versus reported responses.

Our study also contributes to the literature on the economic effects of VAT. Prior research on VAT has focused on tax revenues and compliance (Pomeranz, 2015; De Paula and Scheinkman, 2010; Keen and Lockwood, 2010; Aneja et al., 2021; Hoseini and Briand, 2020), product exports (De Simone and Olbert, 2026), or investments and payouts (Jacob et al., 2019; Jacob et al., 2024). We shift the focus toward the impact of VAT adoption on firms' product mix decisions.

Our study further contributes to the literature at the intersection of development and public economics and to the emerging accounting literature on developing and emerging markets (see, e.g., Hoopes et al., 2023; Aguilar et al., 2025; Chang et al., 2025; Ferguson et al., 2025; Chang, 2026). This literature has emphasized regulation arbitrage, compliance challenges, tax-administration frictions, and information asymmetries that undermine tax revenue collection and economic activity (e.g., Carrillo et al., 2017; Slemrod et al., 2017; Naritomi, 2019; Slemrod, 2019; Kumler et al., 2020; Känzig et al., 2024; Brockmeyer and Sáenz Somarriba, 2025). Prior literature shows that one advantage of VAT is the self-enforcing nature of the invoice-based input tax credit, which helps reduce tax evasion and secure tax revenues (e.g., Keen and Lockwood, 2010; Pomeranz, 2015; Brockmeyer et al., 2024). Our study highlights another advantage of functioning input tax credits: reduced distortions in product mix decisions and higher investment efficiency.

Finally, our study relates to the literature on corporate product mix decisions. Prior studies identify several determinants of product mix, including firm-specific factors (e.g., productivity), product-specific characteristics (e.g., life cycles), industry-level factors (e.g., market structure and competition), and consumer preferences for variety (Bailey and Friedlaender, 1982; Brander and Eaton, 1984; Johnson and Myatt, 2003; Hamilton, 2009; Goldberg et al., 2010b; Bernard et al., 2011; Argente et al., 2024; Kroft et al., 2024). Building on this literature, we offer a new perspective on the tax implications for intra-firm resource adjustments related to product creation and destruction.

Our study has potential policy implications. Because implementation is challenging for (i) a sales tax that taxes only consumers at the final point of sale and (ii) a VAT with immediate and full input tax credits, consumption taxes can impose tax burdens on input goods. Our results imply that these tax burdens adversely affect firms' product mix decisions, reducing profitability and investment efficiency. Moreover, our findings highlight the importance of ensuring full and timely input tax credits under a VAT. Although a VAT can increase compliance burdens, timely and full tax credits can prevent distortions in key business decisions, such as vertical integration. Given the important role of consumption taxes in raising government revenue, understanding these effects is especially critical in developing economies, where full and timely input tax credits are not always granted (IMF, 2021; Brockmeyer et al., 2024).

2. Institutional background and conceptual framework

2.1. Economic development in India in the early 2000s

In the early 2000s, India had a population of approximately 1.1 billion people, making it the world's second-most populous country. Despite its size, India's GDP per capita was about \$800 (in constant 2015 USD), placing it in the global bottom quartile and classifying it as a developing nation. Building on market-oriented reforms begun in the 1990s, central and state governments introduced a series of policies to promote growth. Key among these were the 2005 Special Economic Zones Act, which spurred investment across states, and the 2006 National Rural Employment Guarantee Act—a landmark law rolled out in phases that guarantees 100 days of annual wage employment to rural adults for unskilled manual work. This period also saw rising foreign direct investment and major infrastructure initiatives, such as the completion of the Golden Quadrilateral highway project in 2006–2007, which improved connectivity between key urban centers. Concurrently, however, India faced challenges, including natural disasters such as heavy rains and flooding.

One key driver of the Indian economy is the manufacturing sector, which is the focus of our study. In the early 2000s, manufacturing production facilities were predominantly located in the headquarters' state. In other words, manufacturing companies tended to operate locally rather than across states. This was due to two reasons. First, India's poor physical infrastructure at the time limited the efficiency of cross-state supply chains, leading most manufacturing firms to operate within a single state. As of 2011, only 20% of India's national highway network—which carries 40% of total traffic—consisted of four-lane roads, and one-third of the rural population lacked access to all-weather roads (World Bank, 2014). Second, policy-induced barriers created frictions in interstate trade. India's complex tax system, involving central and state taxes, required checkpoints where officials processed paperwork, inspected trucks, and collected taxes. These checkpoints caused lengthy delays and created opportunities for bribe solicitation. This logistical burden accounted for 0.98% of GDP in 2002 (Das-Gupta, 2006). These frictions led to predominantly within-state sales for Indian firms.

2.2. The VAT reform

Since Indian independence in 1947, the sales tax has been the primary revenue source for Indian states, accounting for about 60% of revenues. One key issue with the sales tax was its cascading nature, meaning that inputs were taxed before the final output was taxed. In a government White Paper on state-level sales taxes (see <https://www.daman.nic.in/notifications/1.pdf>), policymakers note that “[i]n the existing sales tax structure, there are problems of double taxation of commodities[...]. [I]nputs are first taxed, and then after the commodity is produced with input tax load, output is taxed again.” In our sample of firms, we find that firms had to pay sales taxes on

inputs of about 1.3% relative to their total sales. This cascading tax resulted from the difficulty of distinguishing between sales for production purposes and sales for consumption purposes (Hoseini and Briand, 2020). Importantly, this issue was not unique to India. For example, in the U.S., more than 40% of sales tax revenues were paid by firms (Ring Jr, 1999; Bird and Smart, 2014), equivalent to tax payments of 1.16% to 3% of their sales.

Already in the early 1990s, the Indian federal and state governments began considering replacing state-specific sales taxes with a VAT. In the early 2000s, the federal government issued guidelines encouraging states to harmonize their tax schedules and implement a VAT by April 2003. One key objective of the VAT was to eliminate the cascading tax along the supply chain. In the abovementioned White Paper, policymakers state that under a VAT, “a set-off is given for input tax as well as tax paid on previous purchases.” Industry experts similarly confirm that a VAT eliminates cascading sales taxes. For example, a 2005 report by PricewaterhouseCoopers notes that “[t]he input VAT credit can be utilized against the VAT/Central Sales Tax (CST) payable on the sale of goods. It is thus ensured that the cascading effect of taxes is avoided and only the value addition is taxed” (PwC, 2005, p. 36). To account for lower VAT rates relative to sales tax rates for certain items, states were granted flexibility to exempt up to 10 commodities from the VAT base. In addition, the federal government offered partial compensation for any revenue losses incurred by states during the first three years following VAT adoption.

Despite setting April 1, 2003 as the initial start date for the VAT implementation, only the state of Haryana adopted the VAT on that day.⁷ Other states implemented the VAT at different times through 2008: 21 states in 2005, five in 2006, Tamil Nadu and Pondicherry in 2007, and finally Uttar Pradesh in 2008. The timeline of VAT adoption is illustrated in Fig. 1 (see also Online Appendix Table OA1). The delays in implementation were largely driven by a combination of predetermined election cycles and the extensive legislative and administrative changes required for the reform (Ministry of Finance, 2005; Hoseini and Briand, 2020). For example, Tamil Nadu and Uttar Pradesh initially postponed the reform due to uncertainty arising from electoral conflicts between the Congress Party and its political opponents. Importantly, the timing of VAT adoption was unlikely to have been determined by prior growth in state domestic product, revenues, or expenditures (Aneja et al., 2021). To further corroborate that VAT adoption was driven more by political than economic factors, we show that adoption timing is not correlated with state economic characteristics, such as bank credit, government revenues, state GDP, or state-level economic and tax policies aimed at promoting investment (see Online Appendix Table OA2). Consistent with the notion that the timing of the VAT adoption is more politically motivated, we find a significant association with a political factor: states governed by parties in opposition to the federal government adopted the VAT later.

The staggered adoption of the VAT allows us to explore differences in adoption dates across states. Moreover, firms were unlikely to adjust their production plans in anticipation of the reform or engage in intertemporal substitution because the old sales tax regime remained in effect until the VAT was actually implemented. Hence, any change in the product mix is likely to occur only after the adoption of the VAT in the respective state. Under the prior tax regime, statutory sales tax rates varied across states, ranging from 3% to 13%, with most states levying rates between 10% and 12%. Under the new tax regime, the VAT has two basic rates of 4% and 12.5% (Ministry of Finance, 2005; Agrawal and Zimmermann, 2025).⁸ We exploit these differences in our robustness tests to show that our empirical findings do not merely reflect differences in tax rates but rather the change in the tax regime itself.

2.3. Conceptual framework: consumption taxes and firms' decision to vertically integrate

Prior theoretical work on corporate product mix decisions considers supply-side forces, including production costs associated with economies of scale, as well as demand-side forces such as consumer preferences and strategic preemption and interaction among firms in the product market (Bailey and Friedlaender, 1982; Brander and Eaton, 1984; Johnson and Myatt, 2003; Bernard et al., 2011). In this paper, we focus on how consumption taxes affect corporate product mix decisions through vertical integration. Specifically, while in theory consumption taxes should not affect production decisions, we argue that consumption tax systems can distort production decisions when externally sourced upstream inputs are subject to sales taxes. A tax burden on input goods can also arise under a VAT when input tax credits are not fully granted or when they involve substantial delays. We use the switch from a cascading sales tax to a VAT with input tax credits to illustrate the consequences of taxing upstream input goods for a firm's decision to vertically integrate (see Section 2.3.1). In Section 2.3.2, we extend this discussion by incorporating the fact that the impact of consumption taxes on a firm's product mix decisions depends on the tax incidence borne by the firm relative to other stakeholders, such as suppliers or customers (e.g., Weyl and Fabinger, 2013). Finally, in Section 2.3.3, we discuss the role of demand-side effects of consumption taxes on product mix decisions (e.g., Hamilton, 2009; Kroft et al., 2024).

2.3.1. Taxing inputs with sales taxes and firms' decision to vertically integrate

To illustrate how sales taxes can affect vertical integration, that is, the decision to also produce input goods, we consider a simple case in which a profit-maximizing firm can produce an input good I and a consumer good C . The production of C requires the input I . The input good I can be produced internally or sourced externally. If the firm produces the input good I internally, it produces two goods (I and C) and is vertically integrated. Alternatively, the firm can outsource the production of the input good I and produce only

⁷ Facing criticism from the opposing party for being responsible for any flaws in the reform, the Congress Party, which controlled the central government, lifted the April 2003 deadline. The party stated that the VAT would be implemented only once all states were ready, in order to prevent public outrage ahead of the 2004 general election. The April start coincides with the fiscal year for Indian firms, which begins on April 1 and ends on March 31 of the following year.

⁸ State VATs covered 550 commodities, of which about 270 were taxed at the 4% rate. The default sales tax rates in most states were between 8% and 10%. However, they ranged from 3% in Puducherry to 13% in Maharashtra (Agrawal and Zimmermann, 2025).

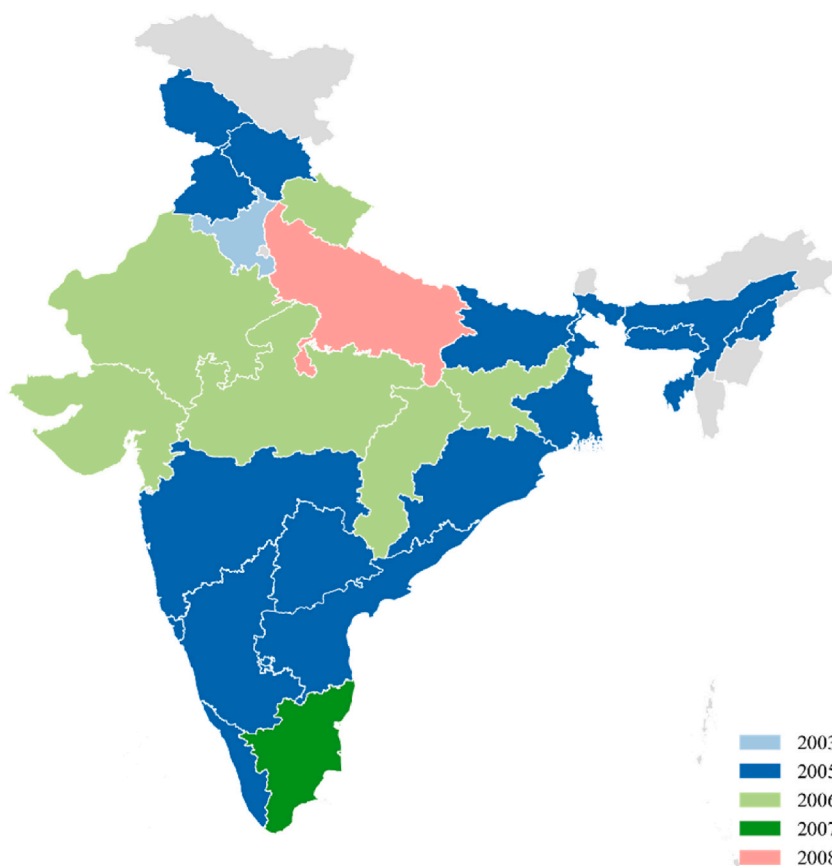


Fig. 1. Map of staggered adoption of VAT across states in India.

This figure illustrates the staggered adoption of VAT across Indian states over the years. Telangana was carved out from Andhra Pradesh in 2014, both of which are shown separately in the map.

the consumer good C . In this case, the firm is not vertically integrated. To illustrate the tax effects, we assume that, absent any taxes, the internal production of the input good I has a negative net present value ($NPV_I < 0$)⁹ and that producing the consumer good C is profitable ($NPV_C > 0$).¹⁰ Further, we assume that the firm realizes a positive overall NPV ($=NPV_I + NPV_C$) irrespective of whether it outsources production ($NPV^{Outsourcing} > 0$) or vertically integrates ($NPV^{VI} > 0$). However, because producing the input good internally is less efficient than sourcing it externally ($NPV_I < 0$), the firm prefers to outsource production absent any tax considerations ($NPV^{Outsourcing} > NPV^{VI}$).

We start with a tax without any double taxation, as illustrated in the numerical example in Panel A of Appendix A. If the sales tax is charged only at the consumer level and there is no tax burden along the supply chain, taxes do not affect the firm's product mix. The sales tax and the VAT lead to the same profit and consumer price. Hence, consumption taxes have no effect on vertical integration. The firm outsources production because $NPV^{Outsourcing} > NPV^{VI}$. The decision to vertically integrate is driven by nontax characteristics, as reflected in the profitability of producing versus outsourcing the input good (NPV_I).

If, however, sales taxes are charged during the production process and not just when selling to the end customer, inputs are subject to double taxation, resulting in a higher total tax burden (see Panel B of Appendix A). This tax burden on externally supplied goods makes outsourcing less profitable. Hence, the internal production of input goods may now become more beneficial than external production ($NPV^{Sales Tax} > 0$), and firms may internally produce the input goods and vertically integrate ($NPV^{Outsourcing, Sales Tax} < NPV^{VI}$).

In contrast, under a VAT system, there is no additional tax burden on inputs because firms receive a tax credit for the VAT on externally sourced inputs. Hence, vertical integration (assuming full and timely tax credits) has no tax-driven benefit. This changes the outcome of the firm's decision to vertically integrate, especially when considering the costs of vertical integration, such as increased coordination costs and the potential loss of business focus. In this case, the firm chooses to produce only the consumption good C and to

⁹ This negative NPV reflects, for example, that external suppliers are more cost-efficient or offer greater flexibility.

¹⁰ If we flip these assumptions—i.e., the input good is profitable ($NPV_I > 0$) and the production of the consumer good might be outsourced ($NPV_C < 0$)—we would still expect the same effect of VAT adoption on vertical integration.

outsource the production of the input good I , leading to a reduced product scope.

Building on these arguments, we can derive a prediction regarding how a transition from a sales tax to a VAT, as in India, affects firms' product mix decisions. A switch from a cascading sales tax to a VAT lowers the effective tax burden (as it avoids the tax on input goods) and thus reduces the incentive for firms to vertically integrate, particularly when the production of one good in the production chain is not profitable on its own (e.g., if $NPV_I < 0$). We thus formulate our main hypothesis as follows:

H1. The transition from a cascading sales tax to a VAT leads to vertical disintegration and thus reduces the number of products.

Vertical integration can resolve the problem of tax cascading, which occurs in the upstream sourcing of inputs. Since a firm's headquarters location determines the applicable sales tax rate, we show in [Appendix B](#) that the key mechanism—vertical disintegration—through which VAT adoption affects the product mix is less driven by the location of downstream customers but by the headquarters' location which determines the tax burden on the upstream suppliers. In [Appendix B](#), we compare one firm that sells in its home state with another firm that sells in a different state with a lower sales tax rate. This example shows that the advantage of vertical integration is very similar for both firms because vertical integration addresses the cascading tax on *upstream* input goods. The customer's location has only a small impact on the decision to vertically integrate versus sourcing inputs externally. Hence, the incentive to vertically integrate is mostly determined by the firm's headquarter state rather than by the customer's location.

2.3.2. The role of tax incidence

The arguments leading to [H1](#) implicitly assume that statutory tax payments equal the tax burden faced by the firm. However, sales taxes can also fall on other stakeholders ([Weyl and Fabinger, 2013](#); [Genakos and Pagliero, 2022](#)). If other stakeholders bear part of the tax burden, firms' responses depend on who ultimately bears the burden (e.g., [Jacob et al., 2019](#); [Jacob and Zerwer, 2024](#)). That is, the extent to which VAT adoption affects product mix depends on the tax incidence of the cascading sales tax burden on the firm versus on suppliers or customers. In the extreme case where firms can pass the entire sales tax burden to other stakeholders (i.e., when suppliers or consumers are perfectly inelastic), statutory sales tax payments on input goods do not affect the decision to vertically integrate, and the firm would not change its product mix in response to a switch from a cascading sales tax to a VAT. However, in practice, neither suppliers nor consumers are fully inelastic. Hence, part of the cascading sales tax burden falls on the firm itself, depending on the relative elasticities of supply and demand. Importantly, there is heterogeneity in these relative elasticities across firms, suppliers, and customers. In our supplemental empirical tests (see [Section 5.1](#)), we explore different measures of tax incidence to capture the idea that the sales tax burden may, for example, be borne by suppliers (through lower input prices) or consumers (through higher sales prices).

2.3.3. The role of changing consumer preferences

While our arguments focus on how shifting from a cascading sales tax to a VAT affects a firm's product mix through vertical integration decisions, a change in the consumption tax can also influence consumer demand. The theoretical literature on how consumption taxes can change consumer preferences is abundant (e.g., [Hamilton, 2009](#); [Kroft et al., 2024](#)). For example, if consumers change their willingness to pay for variety following VAT adoption, firms might adapt their product offerings and alter their product mixes. However, even if such a demand-side effect explains a shift in product offerings to end consumers (i.e., a change in the degree of horizontal integration), it is unlikely to explain a change in the degree of vertical integration. That said, it is possible that a consumer preference-induced change in horizontal integration triggers a change in vertical integration. In our empirical tests in [Section 4.3](#), we address this concern and show that our findings regarding VAT adoption are unlikely to be driven by changes in consumer preferences.

3. Research design

3.1. Data

Our main data source is the Prowess database maintained by the Centre for Monitoring Indian Economy. Prowess provides firm-level identity and financial information for listed and unlisted Indian companies and has been widely used in prior literature ([Gormley et al., 2012](#); [Manchiraju and Rajgopal, 2017](#); [Rajgopal and Tantri, 2023](#)). We focus on listed firms to obtain information on financial statements and stock prices. Prowess also provides panel data on detailed product-level information, including product names and classifications for each firm and year. The data comprise 2971 unique product types across 139 industries, allowing us to calculate the number of unique products a firm produces in a given year (i.e., the firm's product mix). Following [Aghamolla and Li \(2018\)](#), we match a firm's production data to the administrative state based on the firm's registered office address. We exclude observations with missing or negative total assets, borrowing, or market capitalization. Our final stacked sample contains firm-year observations from 3066 unique firms over the period 1995–2016.

Using Prowess data has three main advantages relative to other sources of product-level data, such as company websites ([Boulland et al., 2026](#)), annual reports, or earnings calls ([Ersahin et al., 2024](#)). First, Indian product data are standardized because Indian firms must disclose product-level information in their annual reports under the 1956 Companies Act. Product data are available for 85% of Indian firms in the manufacturing sector and account for 90% of manufacturing outputs and exports ([Goldberg et al., 2010a](#)). Hence,

our data are not subject to major selection issues because of the mandatory reporting requirements.¹¹ Second, our data cover the full product portfolio of a firm. When we aggregate sales by product, the sum matches the firm-level total sales reported in the financial statements. In contrast, in U.S. settings, existing studies primarily rely on text-based product information disclosed in the business description of the financial report (e.g., Hoberg et al., 2014; Babina et al., 2024). However, product information in Item 1 of the 10-Ks relates only to important products, not to all products. Hence, unlike our data, which cover all products—including input goods not for external sale—product information in the 10-Ks or disclosures, such as on company websites, does not capture the complete product portfolio. Third, unlike these alternative sources, we have detailed information on production units, sales units, inventory units, and the sales and inventory values for each product and year for each Indian firm. This allows us to measure unit manufacturing costs and unit sales prices for each product.

3.2. Empirical methodology

To examine the impact of VAT adoption on corporate product mix decisions, we use a stacked DiD approach (e.g., Cengiz et al., 2019; Baker et al., 2022). To construct our final sample, we create individual datasets around each VAT adoption event using all listed firms. For each event, we remove observations that have already been treated. The events and corresponding control regions are listed in Table OA1 of the Online Appendix. We then stack the VAT-wave datasets in relative time to obtain an average treatment effect across all events (Cengiz et al., 2019; Baker et al., 2022). We focus on a $[-8, +8]$ year window around each VAT adoption event e . This allows us to estimate the product mix response to VAT over a longer period, as adjusting corporate product strategy takes time due to sunk entry or adjustment costs (Bernard et al., 2010). We then estimate the following stacked DiD model:

$$Products_{i,s,e,t} = \beta VAT_{s,e,t} + \gamma FirmControls_{i,e,t-1} + \varphi StateControls_{s,e,t-1} + \delta_{i,e} + \lambda_{j,e,t} + \varepsilon_{i,s,e,t}, \quad (1)$$

Where the dependent variable, *Products*, is the number of products by firm i in state s , event e , and year t . *Products* captures the number of product categories (or types). For example, if a car manufacturer produces three different models (e.g., a sedan, an SUV, and a sports car), *Products* equals 1, as these belong to the same category—cars. A higher number of *Products* indicates that the firm produces multiple product categories, such as cars, doors, engines, and belts. Because the number of products may also capture horizontally integrated product categories (e.g., cars and motorbikes), we show in Section 4.3 that our main findings reflect vertical integration.

We use OLS estimates because OLS is generally more consistent in specifications with high-dimensional fixed effects. Moreover, our dependent variable, *Products*, is by definition non-zero, as firms must have at least one product. Hence, the problems with zero-inflated count measures discussed, for example, in Chen and Roth (2024) do not apply in our setting. Importantly, we follow the suggestions in Chen and Roth (2024) and Cohn et al. (2022) and show that our inferences continue to hold when we use a Poisson model that addresses concerns about using OLS for count measures (see Appendix Table A3).

The variable of interest is *VAT*, an indicator that equals 1 when the firm's headquarter state s adopts the VAT in year $t = 0$ and in all subsequent years. The definition of *VAT* is event-specific, as indicated by the index e . *VAT* is defined as the interaction of a *VAT State* indicator (equal to 1 for the treated state s in event e) and a *Post* indicator (equal to 1 for all post-reform years).¹² With this setup, the coefficient β in Equation (1) is the DiD estimate capturing the average change in the number of unique products produced by a treated firm after VAT adoption, relative to control firms located in states that have not yet adopted VAT. Consistent with Hypothesis H1, β is expected to be negative. One potential concern is that defining treatment status based on the headquarters' location could introduce measurement error, as sales taxes are determined by the destination of the seller. However, as discussed in Section 2.3.1 and Appendix B, the incentive to vertically integrate under a cascading sales tax is determined by the firm's headquarter state rather than the location of the customer.

To control for other time-varying firm-specific factors affecting the product mix decision, we include a set of covariates that are likely relevant for corporate cost management and production decisions (e.g., Holzhacker et al., 2015). Controls are lagged by one year and winsorized at the 0.5% and 99.5% levels. We include the natural logarithm of total assets (*Log asset*), the leverage ratio (*Leverage*), sales growth (*Sales growth*), the market-to-book ratio (*MB*), and return on assets (*ROA*) to account for size, financing, growth opportunity, and performance. We also include demand uncertainty, defined as the standard deviation of *ROA* over years $t-6$ to $t-1$ (*SD.ROA*), because it can affect capacity expansion and cost structure decisions (Banker et al., 2014). Finally, we account for capital investment (*INV*) and research and development investment (*R&D*; Bernard et al., 2010). Because two-thirds of our sample firms lack information on R&D, we include the indicator variable *MissingRD*, which equals 1 if R&D information is missing. In this case, we replace missing *R&D* values with 0, following prior literature. Detailed variable definitions are presented in Appendix Table A1.

We also control for several state-level characteristics to account for potential confounding factors that may correlate with VAT adoption and firms' product mix decisions. Based on the events discussed in Section 2.1, we define the following variables. First, the 2005 Special Economic Zones Act is captured by *SEZ*, an indicator for state-years with newly established zone projects. Second, to control for the 2006 National Rural Employment Guarantee Act, we use *NREGA*, an indicator for state-years under the act. Third, to

¹¹ Figure OA1 provides an illustrative example of product information in Prowess using one large motor company. When we checked this product information against the company's past websites using the Wayback Machine (<http://archive.org/>), we found details on the main products sold to customers but no information on the production of intermediary or input products. Hence, the data in our study appear to be more comprehensive.

¹² Because we include event–firm fixed effects and event–year fixed effects, the main effects of *VAT State* and *Post* are absorbed. Only the interaction *VAT State* $\times$ *Post* is identified. For simplicity, we therefore define *VAT* as *VAT State* $\times$ *Post*.

account for increased capital flows, we control for *FDI*, defined as the log of foreign direct investment into state *s* in year *t*. Fourth, differences in political alignment are measured by *Opp_Party*, an indicator for state-years ruled by parties opposed to the federal government; this variable accounts for the delayed VAT adoption in such states. Fifth, we control for the completion of the GQ highway project with *GQ*, an indicator for firms with an entry point within 50 km of the GQ network. *GQ_dist* is a continuous variable based on *GQ*, which replaces *GQ* with the log of distance to the GQ system. Sixth, state-level economic and tax incentives are captured by the indicator *State_econ_inc*. Finally, potential disruptions from natural disasters are measured by *Natural_disaster*, defined as the log of deaths from such events in a state-year. Ideally, we would include state-year fixed effects to account for any unobservable state-year characteristics, but these fixed effects would absorb VAT. Therefore, we use the specific control variables in our baseline approach. When we explore differences across firms, i.e., when we interact VAT with another variable in cross-sectional tests, we also include state-year fixed effects to account for any unobservable state-year characteristics.

We include event-specific firm fixed effects, $d_{i,e}$, to control for time-invariant firm characteristics related to product mix decisions.¹³ We also include event-specific industry-year fixed effects, $l_{j,e,b}$ to absorb economy-wide shocks and time-varying industry-level factors, such as industrial policies or changes in the competitive environment (e.g., Bernard et al., 2010, 2011; Goldberg et al., 2010b). These fixed effects also ensure that we compare treated and control firms within the same industry, thereby controlling for industry-specific nontax incentives to vertically integrate. We cluster standard errors at the state-event level because this is the level at which treatment is identified.

3.3. Descriptive statistics

Table 1 presents descriptive statistics for the main variables used in this study. On average, the number of product categories manufactured by a firm is 4.3, with a standard deviation of 3.8. Recall that this measure captures the number of product categories (e.g., tires, steel parts, and cars) rather than the product range within a category (e.g., different models of a car). Accordingly, more than 80% of firms produce more than one product category. Moreover, approximately 62% of firms are vertically integrated (see Section 4.3 for more details). In our sample, the average firm has total assets of approximately USD 111 million, sales growth of 34.4%, an ROA of 12.1%, and capital investments equal to 6.2% of total assets. These firm characteristics are quantitatively comparable to those reported in Gormley et al. (2012) and Manchiraju and Rajgopal (2017).

4. Main results

4.1. Baseline results: VAT and product mix decisions

Table 2 presents the main results from estimating Equation (1). We include firm and industry-year fixed effects in all columns. Column 1 presents the specification without control variables. In columns 2 and 3, we stagger the inclusion of control variables. Column 4 reports results from the full model with firm- and state-level controls. Consistent with our prediction, the coefficient on VAT is significant and negative, suggesting that firms respond to VAT adoption by reducing their product scope. Notably, the inclusion of state-level controls has little effect on the coefficient estimate, suggesting that the VAT effect on product mix is unlikely to be confounded by observable state-level time-varying factors. In economic terms, the coefficient in column 3 (4) indicates that a firm reduces its number of product categories by 0.13 (0.15). Put differently, approximately one in eight (seven) affected firms reduces its product mix by one category.

To provide additional insight into the economic magnitude and to benchmark our results against the prior literature, we compute the semi-elasticity of the product mix with respect to cascading taxes. Because the cascading tax burden of vertically integrated firms is not observable—firms vertically integrate to avoid paying sales taxes on input goods—we calculate a hypothetical cascading sales tax burden by simulating the case in which a firm sources all inputs externally. The results of this estimation are reported in Appendix Table A2. We find a semi-elasticity of the product mix with respect to the sales tax of approximately -0.8 to -1 . This semi-elasticity is lower than those reported in prior studies on VAT effects (e.g., De Simone and Olbert, 2026), but it remains economically meaningful. Overall, these results show that, consistent with H1, firms subject to VAT adoption adjust their product strategy by reducing product scope and adopting a more focused product strategy.

We further subject our main findings to several robustness tests, which are reported in Appendix Table A3. First, we examine whether the reduction in the number of products is driven by discontinuing existing products or by adding fewer new products. To do so, we construct *Proportion of Continued Products*, defined as the number of unique products that continue to be manufactured in subsequent years divided by the total number of unique products manufactured at the beginning of each event window. We find evidence that firms reduce product categories after VAT adoption (see column 1). Second, we address the concern that treatment intensity may vary when firms operate in multiple states. Accordingly, we restrict the sample to firms operating in a single state. Our inferences remain unchanged (column 2).¹⁴ Third, our results are robust to replacing the dependent variable *Products* with its natural logarithm (column 3). Fourth, we find similar results when we use a Poisson estimator to address concerns about using OLS for count

¹³ We denote VAT event-specific (or wave-specific) firm fixed effects as firm fixed effects throughout the study.

¹⁴ In an untabulated analysis, we replace VAT with a weighted measure, *Weighted VAT*, which uses subsidiary locations to approximate a firm's average exposure to VAT implementation when it operates across states. We find a significantly negative coefficient on *Weighted VAT*, supporting our main findings.

Table 1

Descriptive Statistics

This table presents descriptive statistics for firm-level variables used in the main analysis. Appendix Table A1 presents variable definitions.

Variable	N	Mean	SD	p25	p50	p75
<i>Products</i>	41,214	4.267	3.825	2.000	3.000	5.000
<i>Log_Products</i>	41,214	1.161	0.753	0.693	1.099	1.609
<i>Firm_VI</i>	39,553	0.288	0.336	0.000	0.200	0.500
<i>Firm_VI_Weighted</i>	39,553	0.775	2.231	0.000	0.019	0.629
<i>Dum_VI</i>	37,596	0.622	0.485	0.000	1.000	1.000
<i>Outsourcing</i>	25,395	0.044	0.138	0.000	0.000	0.000
<i>No. of Subsidiaries</i>	12,000	0.728	0.818	0.000	0.693	1.100
<i>Log_asset</i>	41,214	6.838	1.745	5.510	6.681	7.957
<i>Leverage</i>	41,214	0.438	0.231	0.270	0.452	0.611
<i>Sales_growth</i>	41,214	0.343	2.230	-0.024	0.114	0.279
<i>ROA</i>	41,214	0.121	0.089	0.073	0.118	0.163
<i>MB</i>	41,214	1.419	2.360	0.348	0.703	1.506
<i>RD</i>	41,214	0.002	0.006	0.000	0.000	0.001
<i>MissingRD</i>	41,214	0.674	0.469	0.000	1.000	1.000
<i>Investment</i>	41,214	0.062	0.077	0.011	0.035	0.082
<i>SD_ROA</i>	41,214	0.049	0.064	0.021	0.035	0.058
<i>SEZ</i>	41,214	0.184	0.388	0.000	0.000	0.000
<i>NREGA</i>	41,214	0.223	0.416	0.000	0.000	0.000
<i>FDI</i>	41,214	8.563	3.409	8.105	9.628	10.580
<i>Opp_party</i>	41,214	0.490	0.500	0.000	0.000	1.000
<i>GQ</i>	41,214	0.222	0.415	0.000	0.000	0.000
<i>GQ_dist</i>	41,214	2.836	4.492	0.000	0.000	7.471
<i>State_econ_inc</i>	41,214	0.008	0.089	0.000	0.000	0.000
<i>Natural_disaster</i>	41,214	7.126	1.495	6.184	7.085	7.862

variables (column 4). Fifth, column 5 addresses concerns that early and late adoption events may have differential effects. We find that the effect of VAT adoption on product mix is present for early and late adopters. Finally, we address the concern that the effect of VAT adoption on product mix is driven by lower nominal VAT rates relative to sales tax rates. To this end, we drop states where the post-reform VAT rate is lower than the sales tax rate. The results in column 6 show that our inferences remain unchanged.

Next, we address concerns about using the headquarters' location when defining VAT. As discussed in Section 2, vertical integration addresses tax cascading in the *upstream* sourcing of input goods, for which the headquarters' location is decisive in determining the applicable sales tax rate. Hence, the incentive to vertically integrate under a sales tax regime should be determined by the headquarters' state. To provide empirical support for this notion, we examine whether the effect of VAT adoption on the product mix is driven by the tax burden on upstream inputs (i.e., the tax burden determined at the headquarters) or by the tax burden on customers. Because our sample lacks data on customer location within India, we use exports to firms outside India, which are exempt from VAT. If the customer tax burden matters, firms with higher export intensity should exhibit a different change in their product mix than firms with lower export intensity. We therefore interact VAT with *High_Export*, an indicator equal to one if the firm has an above-median level of export intensity. The insignificant VAT $\times$ *High_Export* coefficients in Panel A of Table 3 indicate that the effect of VAT adoption on the product mix does not depend on export intensity. Firms with both high and low export intensities respond similarly to VAT by reducing their product mix. These findings corroborate the argument that the decisive friction is the sales tax in the upstream supply chain, which is determined by the headquarters' state.

4.2. Dynamic effects and spillover effects

A key identification assumption in a stacked DiD design is that, absent the VAT reform, treated and control firms follow similar pre-trends in their product mix. Although this assumption is not directly testable, we examine the dynamic effects around the reform by expanding Equation (1). Specifically, we include VAT(-6), VAT(-5), VAT(-4), VAT(-3), and VAT(-2), which indicate six and preceding years, five, four, three, and two years, respectively, prior to the VAT adoption. We also include an indicator for the reform year, VAT(0), as well as VAT(1), VAT(2), VAT(3), VAT(4), VAT(5), and VAT(6+), which are dummy variables indicating one, two, three, four, five, and six or more years after the VAT reform, respectively. The year prior to reform, (VAT(-1)), serves as the benchmark period and is therefore omitted.

Fig. 2 plots the VAT coefficients. The statistically insignificant coefficients in the pre-VAT period indicate that the difference in the number of products between treated and control firms prior to VAT adoption is indistinguishable from zero. This supports the parallel pre-trend assumption, suggesting that treated and control firms followed a similar trajectory in product mix before the reform. In contrast, firms begin producing fewer product types one year after the VAT adoption in their home states, as reflected in the statistically and economically negative coefficients in the post-VAT period. Consistent with the notion that vertical disintegration requires

Table 2

Value Added Tax and Product Mix Decisions: Baseline Results

This table presents the main results for the effect of VAT on firm product mix decisions estimated based on a stacked DiD regression. The key independent variable is *VAT*, which is an indicator variable that equals one for the year in which an Indian state adopted VAT and for all subsequent years, and zero otherwise. The dependent variable is *Products*, which is the number of unique products a firm produces in a year. [Appendix Table A1](#) presents variable definitions. Standard errors (reported in parentheses) are clustered at the state-event level. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively.

Dependent variable	(1) <i>Products</i>	(2) <i>Products</i>	(3) <i>Products</i>	(4) <i>Products</i>
VAT	−0.177*** (0.051)	−0.129*** (0.048)	−0.133*** (0.045)	−0.149*** (0.050)
<i>Log_asset</i>		0.534*** (0.058)	0.532*** (0.056)	0.538*** (0.058)
<i>Leverage</i>		0.286** (0.112)	0.323*** (0.106)	0.333*** (0.114)
<i>Sales_growth</i>		0.002 (0.003)	0.003 (0.003)	0.002 (0.003)
<i>ROA</i>		−0.287* (0.151)	−0.283** (0.140)	−0.258* (0.147)
<i>MB</i>			−0.008 (0.008)	−0.008 (0.009)
<i>RD</i>			9.970** (3.912)	10.358*** (3.859)
<i>MissingRD</i>			−0.128** (0.053)	−0.120** (0.055)
<i>Investment</i>			−0.391** (0.189)	−0.428** (0.182)
<i>SD_ROA</i>			0.023 (0.201)	0.061 (0.198)
<i>SEZ</i>				−0.075 (0.047)
<i>NREGA</i>				0.090 (0.155)
<i>FDI</i>				0.004 (0.005)
<i>Opp_party</i>				−0.009 (0.025)
<i>GQ</i>				−0.029 (0.194)
<i>GQ_dist</i>				0.078* (0.041)
<i>State_econ_inc</i>				0.218 (0.200)
<i>Natural_disaster</i>				0.021 (0.014)
Industry × Year FE	Y	Y	Y	Y
Firm FE	Y	Y	Y	Y
Observations	42,180	42,180	42,180	41,214
R-squared	0.851	0.854	0.854	0.856

time due to adjustment costs, the negative impact of VAT on the product mix increases over time. The coefficients become statistically insignificant after six years, likely due to sample attrition. The estimates appear to stabilize approximately four to five years after VAT adoption.

In addition to the parallel-trends assumption, we test whether the stable unit treatment value assumption is violated, for example, due to spillover effects. Such effects could occur if changes in vertical integration by treated firms in one state in response to VAT adoption influence control firms operating in the same industry in other states. To examine this, we follow the approach of [Berg et al. \(2021\)](#), as applied by [Hoopes et al. \(2023\)](#), estimating the VAT adoption effect on treated and control firms conditional on treatment intensity. The results of this spillover test are reported in Panel B of [Table 3](#). We find no evidence of spillover effects on other treated firms or on control firms in other states. Importantly, the main coefficient on *VAT* remains negative and significant. These findings indicate that (i) our main estimates are unlikely to be biased by spillover effects, and (ii) there are no spillover effects on control firms.

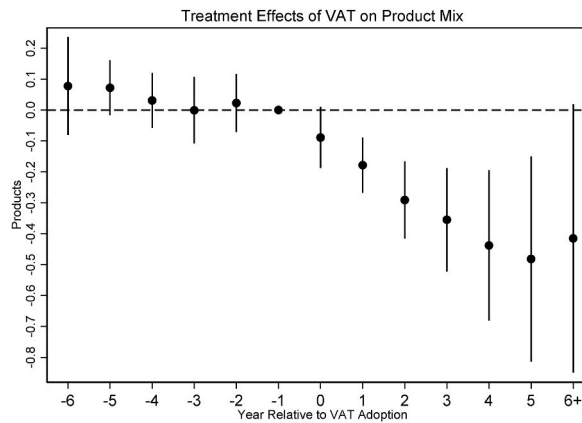
Table 3

Supplemental Analyses

Panel A reports the results of the effect of the VAT on product mix conditional on the reliance of export. *High_Export* is an indicator for firms which generate a larger-than-median level of export intensity. Export intensity is income from exports scaled by total income. Panel B reports the results of potential spillover effects to other firms in the same industry but located in states not yet affected. *IndTreatmentShare* is the proportion of treated firms in an industry, which captures within-industry treatment intensity. Standard errors (reported in parentheses) are clustered at the state-event level. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively.

Panel A. Exporters		
	(1)	(2)
Dependent variable	<i>Products</i>	<i>Products</i>
VAT	−0.160* (0.086)	–
VAT × <i>High_Export</i>	0.017 (0.141)	0.030 (0.139)
<i>High_Export</i> × <i>Post</i>	0.006 (0.048)	−0.004 (0.039)
Controls	Y	Y
Industry × Year FE	Y	Y
Firm FE	Y	Y
State × Year FE	N	Y
Observations	40,565	40,565
R-squared	0.856	0.856

Panel B. Spillover effects in product market		
	(1)	(2)
Dependent variable	<i>Products</i>	<i>Products</i>
VAT	−0.251*** (0.077)	−0.196*** (0.070)
VAT × <i>IndTreatmentShare</i>	0.272 (0.275)	0.281 (0.263)
(1 − VAT) × <i>IndTreatmentShare</i>	0.154 (0.360)	0.175 (0.370)
Controls	N	Y
Year FE	Y	Y
Firm FE	Y	Y
Observations	41,214	41,214
R-squared	0.845	0.850

**Fig. 2.** Dynamic effect of VAT on product mix decision.

This figure plots the coefficients on $VAT(\tau)$ and the 95% confidence interval estimated from the following equation: $Products_{i,s,e,t} = \sum \beta_{(\tau)} VAT(\tau)_{s,e,t} + \gamma X_{i,e,t-1} + \delta_{i,e} + \lambda_{j,t} + \varepsilon_{i,s,e,t}$. $\tau(-6, -5, -4, -3, -2, -1)$ refers to six and preceding years, five years, four years, three years, two years, and one year, respectively prior to the VAT adoption. $\tau(0, 1, 2, 3, 4, 5, 6+)$ refers to zero year, one year, two years, three years, four years, five years, six and subsequent years, respectively after the VAT adoption. The year prior to the VAT adoption (-1) serves as the benchmark period and is thus omitted.

4.3. Mechanism tests: vertical disintegration

Next, we shed more light on the main interpretation—*vertical disintegration*—of the finding that the VAT adoption reduces corporate product offerings. One concern with our main approach is that the observed reduction in the number of products could also reflect changes in horizontally integrated product types driven by shifts in consumer preferences. To address this concern and link our results to vertical disintegration, we leverage the detailed product descriptions in our data to construct a firm-level measure of vertical integration, *Firm_VI*. A key advantage of the product description data from Prowess is that it is disclosed in accordance with the 1956 Companies Act and is therefore standardized, making product information comparable across firms. Moreover, this uniformity facilitates more consistent and accurate classification across firms. Consequently, the input data, i.e., the standardized product descriptions, are less susceptible to the non-standardized language often found in company websites or earnings call disclosures.

Firm_VI is calculated in three steps. Because manually classifying the input–output relationships among over 8.8 million product pairs from the 2971 products in our data is infeasible, we first use GPT-4o-mini to classify whether each pair of products is in an input–output relationship. For each product *A* in a pair, GPT-4o-mini classifies whether product *B* is upstream, downstream, or irrelevant. We run this process for all 8.8 million product pairs. The corresponding prompt and examples of the classification are provided in the Online Appendix, Part B. For instance, belts are an input product for cars, whereas bakery machines are not. Next, we identify the five main products of each firm based on product-level sales.¹⁵ For each main product, we calculate the extent to which all other product types are vertically related to that product. We then divide the number of vertically related products by the total number of non-main products. For example, suppose Firm *M* has six products. Product *A* is one of the main products. Products *B* and *C* are inputs for Product *A*, while Products *D*, *E*, and *F* are not inputs for Product *A*. In this case, we calculate the degree of vertical integration for Firm *M* with respect to its Product *A* as $2/5 (=0.4)$. We repeat this calculation for all main products and average the ratios to obtain the firm-level vertical integration measure, *Firm_VI*. In our sample, approximately 29% of offered products are vertically integrated, and about 62% of firms exhibit some degree of vertical integration prior to their respective VAT adoption.¹⁶

We then use our firm-level vertical integration measure, *Firm_VI*, to directly examine the effect of VAT adoption on vertical integration. We report the results in Panel A of Table 4. In column 1, we find negative coefficients on VAT, indicating a lower degree of vertical integration following the VAT adoption. To account for the relative importance of different products, we also construct a sales-volume-weighted measure, *Firm_VI Weighted*, where weights (in %) are defined as the proportion of each product's sales volume relative to total sales. Column 2 of Panel A shows robust evidence of decreased vertical integration using this weighted measure. Overall, the VAT adoption has an economically meaningful impact on vertical (dis)integration, with an 8.3% (14.5%) decrease in the degree of vertical integration relative to the average values of *Firm_VI* (*Firm_VI Weighted*).

One potential concern is the accuracy of classifying input–output relationships using GPT. To address this, we manually reviewed over 40,000 product pairs to ensure the reliability of the model.¹⁷ GPT achieved an F1 score of 91.16% and an accuracy rate of 85.27% (see Online Appendix, Part B, for details), indicating strong performance for this relational classification task (Yoo, 2024). Second, we used GPT to generate a confidence score for each classification and constructed confidence-score-weighted measures of vertical integration, acknowledging the limitations of GPT-based confidence scores (Bernard et al., 2025). In columns 3 and 4 of Panel A, we show the results using these weighted measures of *Firm_VI* and *Firm_VI Weighted*. We again find that the VAT adoption reduced vertical integration. Importantly, the economic magnitudes remain very similar when we use confidence scores as weights.

Second, we return to our original design and use *Products* as the dependent variable. Our hypothesis predicts that firms that are vertically integrated should respond to the VAT adoption, whereas horizontally integrated firms should not. To test this, we construct an indicator variable, *Dum_VI*, equal to 1 for firms producing any vertically integrated products prior to the VAT adoption. We then rerun Equation (1), including VAT, the interaction of VAT with *Dum_VI*, and the interaction of *Dum_VI* with *Post*.¹⁸ All other interactions or main effects of these variables are absorbed by firm or industry-year fixed effects.

Column 1 of Panel B, Table 4 presents the results. We find that the VAT coefficient—capturing the response of horizontally integrated firms—is insignificant and close to zero, consistent with the notion that horizontally integrated firms should not respond to the VAT adoption. Importantly, this test addresses concerns about alternative explanations, such as changes in consumer preferences around VAT adoption, which would induce firms to introduce new product categories (e.g., Hamilton, 2009). Because any demand-side effects would similarly affect vertically and horizontally integrated firms, the insignificant response among horizontally integrated firms indicates that our main results are unlikely to be driven by changes in consumer preferences or other factors affecting both firm types. Importantly, we find significant and negative coefficients on $VAT \times Dum_VI$, as well as significantly negative joint

¹⁵ We select five products because the average firm has 4.3 products, and firms at the 75th percentile have five products.

¹⁶ To further validate *Firm_VI*, we calculate the average pre-VAT effective sales tax rates—defined as firms' sales tax payments scaled by total sales—for horizontally and vertically integrated firms. The effective tax rate is 1.45% for the former, while that is 1.21% for the latter. The difference between the two groups is statistically significant (t-stat = 1.85). This shows that by vertically integrating, firms were able to lower sales taxes on input goods.

¹⁷ To manually verify the LLM predictions, we assigned each product pair to three independent research assistants, who reviewed the inputs, generated their own predictions (using majority voting or discussion to reconcile discrepancies), and evaluated the LLM's output using two standard metrics: F1 score and accuracy.

¹⁸ In all tests examining variation in the effect of VAT adoption across firms, VAT is defined as the interaction between *VAT State* and *Post*. Because all cross-sectional split variables (e.g., *Dum_VI*) are time-invariant, their main effects and interactions with *VAT State* are absorbed by the firm fixed effects.

Table 4**VAT and Vertical Disintegration**

This table presents direct evidence on vertical disintegration after the VAT adoption (Panel A) and whether the main results—a decrease in product mix—are mainly driven by ex ante vertically integrated firms (Panel B). In Panel A, *Firm_VI* (*Firm_VI_Weighted*) is the degree of firm-year (weighted) vertical integration. We calculate firm-level integration, *Firm_VI*, in three steps. First, we use GPT-4o-mini to identify the input–output relationship between any two products for the universe of products in the database, which is illustrated in Online Appendix, Part B. We end up with product pairwise data. For every product A in the product pair, we identify product B as either upstream or downstream or irrelevant. Second, we identify the five main products of any given firm, defined based on firm-product sales. Third, for each one of the main products of a firm, we obtain the extent to which other non-main products are vertically related to the main product by calculating the ratio of vertically related products to total number of non-main products. *Firm_VI* is then calculated as the average value of the ratios with respect to those main products. We also define a sales-volume-weighted measure of firm-level vertical integration, *Firm_VI_Weighted*, with weights (in %) being the proportion of sales volume of each of the products relative to total sales. In columns (3) and (4) of Panel A, we weight the *Firm_VI* and *Firm_VI_weighted* by the confidence scores generated by GPT in identifying input-output relationships. In Panel B, *Dum_VI* is an indicator variable for firms that were vertically integrated prior to the VAT adoption. Standard errors (reported in parentheses) are clustered at the state-event level. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively.

Panel A. Firm-level vertical integration				
	(1)	(2)	(3)	(4)
	<i>Not weighted by confidence score</i>	<i>Not Weighted by confidence score</i>	<i>Weighted by confidence score</i>	<i>Weighted by confidence score</i>
Dependent Variable	<i>Firm_VI</i>	<i>Firm_VI_Weighted</i>	<i>Firm_VI</i>	<i>Firm_VI_Weighted</i>
VAT	−0.024*** (0.007)	−0.112*** (0.029)	−0.022*** (0.006)	−0.092*** (0.025)
Controls	Y	Y	Y	Y
Industry × Year FE	Y	Y	Y	Y
Firm FE	Y	Y	Y	Y
Observations	39,553	39,553	39,553	39,553
R-squared	0.778	0.861	0.776	0.859
Panel B. Heterogeneity by vertical integration prior to the VAT adoption				
	(1)	(2)		
Dependent variable	<i>Products</i>	<i>Products</i>		
VAT (i)	0.038 (0.097)	–		
VAT × <i>Dum_VI</i> (ii)	−0.270** (0.110)	−0.251** (0.106)		
<i>Dum_VI</i> × <i>Post</i>	−0.021 (0.041)	−0.028 (0.031)		
<i>Joint Significance</i> (i) + (ii) <i>[t-stat]</i>	−0.232*** <i>[3.53]</i>	−0.251** <i>[2.36]</i>		
Controls	Y	Y		
Industry × Year FE	Y	Y		
Firm FE	Y	Y		
State × Year FE	N	Y		
Observations	37,596	37,596		
R-squared	0.856	0.856		

effects of VAT and $VAT \times Dum_VI$.

In column 2, we include state-year fixed effects to account for any time-varying state-level factors. This more stringent specification mitigates concerns over unobservable state-level confounders, complementing our baseline approach, which controls for observable state-level factors. By design, the coefficient on VAT is absorbed by the state-year fixed effects. We find that the coefficient on $VAT \times Dum_VI$ remains negative and significant, indicating that, relative to horizontally integrated firms (the unaffected control group), vertically integrated firms reduce the number of products. Importantly, the inclusion of state-year fixed effects does not change the coefficient estimates, suggesting that our findings are unlikely driven by unobservable state-level factors. In economic terms, the results in Panel B indicate that about one in four firms reduce a product type following VAT adoption. Overall, these results show that, after the VAT adoption, firms decrease the number of product types they produce and that this reduction is driven by vertically integrated firms, consistent with vertical disintegration following the reform.¹⁹

¹⁹ To corroborate that VAT adoption narrows product scope through vertical disintegration, we examine three additional predicted consequences: reduced firm size, increased outsourcing, and subsidiary divestment. As shown in Online Appendix Table OA3, firms subject to VAT adoption exhibit a statistically significant reduction in size relative to the control group, consistent with disintegration. Second, we observe increased reliance on external production, indicating greater outsourcing. Third, the number of subsidiaries declines significantly, further supporting the vertical disintegration interpretation.

4.4. VAT and product specialization

4.4.1. Within-firm shift to profitable products

In this section, we provide additional insights into product mix adjustments in response to the VAT adoption by examining the following question: given that firms become more focused in their product mix strategy, do firms specialize in certain product types, and if so, in which products do they specialize? The objective of this test is to link our findings more to the prediction that firms outsource the production of the least profitable projects, which we illustrate via the input good I with a negative NPV ($NPV_I < 0$).

We investigate this question in two steps. First, we examine whether firms specialize in certain products by constructing a within-firm product concentration measure (*Product Concentration*). This measure draws on the literature on industry concentration, which uses the Herfindahl–Hirschman (*HHI*) index to assess market concentration (e.g., Grullon et al., 2019). We apply the *HHI* logic to a firm's product portfolio: *Product Concentration* is calculated as the sum of the squared revenue shares of all products produced by the firm in a given year, where each product's share is its revenue divided by the firm's total revenue.²⁰ Higher values of *Product Concentration* indicate greater specialization in specific products. Second, we construct a profitability-weighted measure of product concentration (*Markup-Weighted Product Concentration*). Here, we calculate the weighted sum of squared product shares, using the product-level markup—defined as sales minus cost, divided by cost—as weights. A higher *Markup-Weighted Product Concentration* indicates that the firm focuses on producing its more profitable products.

We then re-estimate Equation (1) using these two measures and present the results in Panel A of Table 5. In columns 1 and 2, we find positive and significant coefficients on VAT when using *Product Concentration* as the dependent variable. This suggests that firms adopt a more concentrated product strategy with less product diversity following the introduction of VAT. This finding corroborates our results in Table 2. In columns 3 and 4, we explore whether firms focus on products with the highest markups using *Markup-Weighted Product Concentration* as the dependent variable. Consistent with our arguments in Section 2, we again find positive and significant coefficients on VAT. Notably, the coefficients on VAT in columns 3 and 4 are nearly 10 times larger than those in columns 1 and 2.²¹ These findings suggest that, in response to VAT adoption, firms adjust their product mix by reducing existing products and reallocating resources toward fewer, more profitable products.²²

4.4.2. Expansion of subtypes of profitable products

Our findings so far suggest that firms respond to the VAT adoption by *reducing* their product category scope while retaining more profitable products. In other words, firms adopt a more specialized product strategy. In this section, we zoom in on firms' product specialization strategies by examining whether firms subject to VAT expand the number of products *within* the categories on which they now concentrate. Recall that one product category (e.g., cars) can have multiple subtypes (e.g., sedan, SUV, or sports car). We next test whether firms, while cutting product categories (e.g., discontinuing tires), expand offerings within key product subtypes (e.g., introducing a new car model). To do so, we define *Within-Type Products* as the number of subtypes a firm produces within a given product category in a year, and use this as the dependent variable. We estimate Equation (1) at the product-type level, which provides many more observations and allows for inclusion of more granular fixed effects at the product category level. The results are reported in Panel B of Table 5. The coefficient on VAT in column 1 is positive and statistically significant, indicating that firms expand subtypes within a product category after VAT adoption (i.e., they specialize more in a product category). The point estimate of 0.036 corresponds to a 3% increase in the number of product subtypes relative to the sample mean.

Next, we examine this specialization decision in more detail by investigating whether the increase in within-category products is driven by the most profitable products. To this end, we define *High Markup Products*, an indicator for above-median markups within a product category and industry. We then interact *High Markup Product* with VAT and with *Post*, expanding the baseline model. The results are reported in columns 2–4 of Panel B, Table 5, where we gradually introduce more stringent fixed effects. In column 2, we include firm, industry-year, and product-type fixed effects. In column 3, we add state-year fixed effects to absorb any unobservable state-year characteristics. In column 4, we further exploit variation within firm-product categories by controlling for firm-product category fixed effects. Importantly, the coefficients on $VAT \times High\ Markup\ Products$ are positive and significant across all specifications. The economic magnitudes are similar across columns, indicating that unobservable state-year characteristics have little impact on the estimate of the interaction term. Collectively, these results suggest that firms specialize in their most profitable product types. While firms reduce the number of product types (i.e., vertical disintegration), they expand offerings *within* the most profitable products. This suggests that firms redirect resources previously tied to less-profitable products to expand the production of higher-margin products.

²⁰ For example, suppose a firm has four unique products with revenues of $Sales_1$, $Sales_2$, $Sales_3$, and $Sales_4$, respectively. The product portfolio share for Product 1 is calculated as $Share_1 = Sales_1 / (Sales_1 + Sales_2 + Sales_3 + Sales_4)$. *Product Concentration* is then computed as $Share_1^2 + Share_2^2 + Share_3^2 + Share_4^2$.

²¹ The average value for *Product Concentration* (*Markup-Weighted Product Concentration*) is 0.73 (0.45).

²² In Online Appendix Figure OA2, we plot the dynamic effects of VAT on vertical integration, product concentration, and markup-weighted product concentration. The figure shows parallel trends in these variables before VAT implementation and divergent patterns following the reform.

Table 5**VAT and Product Specialization**

This table presents results on the extent to which the VAT incentivizes product specialization. In Panel A, *Product Concentration* in columns (1) and (2) is defined as the sum of the squared portfolio shares for all products a firm produces in a year, with a higher (smaller) value indicating that products are more concentrated (diversified). Product portfolio share is calculated as the ratio of revenue generated from each product of a firm to the total revenues generated from all products of the firm. *Markup-Weighted Product Concentration* in columns (3) and (4) is defined similarly as *Product Concentration*, except that the squared portfolio shares of different products a firm produces in a year are weighted by the markup of each product. *Markup* is defined as sales minus cost divided by cost for each product. In Panel B, we report whether VAT leads to a more focused product strategy by producing more *subtypes* of products, while reducing the number of unique *types* of products. The dependent variable is *Within-Type Products*, which is the number of subtypes of products for a given firm-product type in a year. *High Markup Products* is an indicator for products that have higher-than-median markups within a product type and an industry. The analysis is conducted at firm-product type-year level. Standard errors (reported in parentheses) are clustered at the state-event level. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively.

Panel A. Within-firm shifting to high-markup products				
	(1)	(2)	(3)	(4)
Dependent variable	<i>Product Concentration</i>		<i>Markup-Weighted Product Concentration</i>	
VAT	0.026*** (0.007)	0.023*** (0.006)	0.220*** (0.039)	0.180*** (0.043)
Controls	N	Y	N	Y
Industry × Year FE	Y	Y	Y	Y
Firm FE	Y	Y	Y	Y
Observations	41,165	41,165	26,717	26,717
R-squared	0.664	0.666	0.370	0.371
Panel B. Expanding to subtypes of products that are more profitable				
	(1)	(2)	(3)	(4)
Dependent variable	<i>Within-Type Products</i>	<i>Within-Type Products</i>	<i>Within-Type Products</i>	<i>Within-Type Products</i>
VAT (i)	0.036*** (0.013)	0.020 (0.017)	–	–
VAT × High Markup Products (ii)		0.061** (0.027)	0.064** (0.026)	0.051* (0.030)
High Markup Products × Post		0.005 (0.012)	0.003 (0.012)	–0.010 (0.020)
Joint Significance (i) + (ii) [t-stat]		0.081*** [3.64]	0.064** [2.4]	0.051* [1.73]
Controls	Y	Y	Y	Y
Industry × Year FE	Y	Y	Y	Y
Firm FE	Y	Y	Y	N
Product Type FE	Y	Y	Y	N
Firm × Product Type FE	N	N	N	Y
State × Year FE	N	N	Y	Y
Observations	175,861	175,861	175,861	175,861
R-squared	0.385	0.386	0.386	0.690

5. Heterogeneity tests

5.1. Heterogeneity by tax incidence

In this section, we explore one important source of heterogeneity in firms' product mix responses to VAT adoption: the role of tax incidence. Firms are likely to respond more strongly to the VAT adoption if they bear a larger share of the consumption tax burden, and less strongly if suppliers or customers bear more of the burden. In other words, when firms can pass a greater portion of the cascading sales tax on inputs to other stakeholders, the decision to outsource or produce certain products internally is less affected by taxes, making changes in the product mix less likely. We test this prediction in two sets of tests—one exploring product market concentration and product markups (Section 5.1.1) and the other exploring differences in tax incidence across firms (Section 5.1.2).

5.1.1. Product market concentration and product markups

The first set of tests builds on theoretical and empirical work showing that firms in more concentrated markets pass on a smaller share of the tax burden to consumers, whereas firms in more competitive markets pass on more when maximizing profits (e.g., [Weyl](#)

Table 6**Heterogeneity by Product Market Concentration and Product Markups**

This table reports the results of the effect of VAT on product mix conditional on product market concentration (Panel A) and product markups (Panel B). In Panel A, we use *Product HHI* to proxy for the product market concentration. To construct the measure, we first estimate the *HHI* of each product, which is the sum of squared market shares of all firms in a given product market. Multiple-product firms with n number of products thus have n number of *HHI*. Then, for each firm, we obtain the standardized average value of *HHI*, *Product HHI*, for the period prior to the tax reform. A larger value of *Product HHI* indicates a more concentrated market faced by a firm. In Panel B, *Firm Markup* is constructed as follows. We first estimate the markup of each product produced by a firm, which is defined as sales minus cost divided by cost for each product. Multiple-product firms with n number of products thus have n number of product markups. To make the markups in different product types comparable, we normalize the markups to get the relative position of a firm's product in the whole product market by subtracting the minimum value of product-market-wide markups and dividing by the range between the maximum and minimum values of product-market-wide markups. Then, for each firm, we obtain the average normalized markups, *Firm Markup*, for the pre-reform period. A larger value of *Firm Markup* indicates a higher average product markup by a firm. [Appendix Table A1](#) presents variable definitions. Standard errors (reported in parentheses) are clustered at the state-event level. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively.

Panel A. Heterogeneity by product market concentration		
	(1)	(2)
Dependent variable	<i>Products</i>	<i>Products</i>
VAT	−0.145** (0.060)	–
VAT × <i>Product HHI</i>	−0.147*** (0.054)	−0.155*** (0.054)
<i>Product HHI</i> × Post	−0.000 (0.106)	−0.026 (0.077)
Controls	Y	Y
Industry × Year FE	Y	Y
Firm FE	Y	Y
State × Year FE	N	Y
Observations	40,184	40,184
R-squared	0.856	0.857
Panel B. Heterogeneity by average product markup		
	(1)	(2)
Dependent variable	<i>Products</i>	<i>Products</i>
VAT	−0.200*** (0.071)	–
VAT × <i>Firm Markup</i>	0.054* (0.031)	0.053* (0.032)
<i>Firm Markup</i> × Post	0.115*** (0.025)	0.105*** (0.025)
Controls	Y	Y
Industry × Year FE	Y	Y
Firm FE	Y	Y
State × Year FE	N	Y
Observations	31,665	31,665
R-squared	0.857	0.858

and Fabinger, 2013; Genakos and Pagliero, 2022).²³ This implies that a reduction in product scope is more likely for firms in concentrated markets, which bear a larger share of the tax burden, and less likely for firms in competitive markets, which bear a smaller share. In other words, we expect the VAT to have a larger effect on product mix in concentrated markets and a smaller effect in more competitive ones.

In Panel A of [Table 6](#), we test this prediction using product market concentration (*Product HHI*) as a measure of competition. To construct this measure, we first calculate the *HHI* for each product, defined as the sum of the squared market shares of all firms in that product market. Multiple-product firms with n products thus have n values of *HHI*. Our final measure, *Product HHI*, is the average *HHI*

²³ Intuitively, in highly competitive markets, firms respond to sales taxes primarily by reducing output, which raises prices and passes most of the tax burden onto consumers. In contrast, in more concentrated markets, where firms equate marginal revenue with marginal cost, adjustments in quantity and price are smaller, so firms bear a larger share of the tax burden.

Table 7

Heterogeneity by Firm-level Tax Incidence

This table reports the results of the effect of the VAT on product mix conditional on firm-level sales tax incidence on customers and suppliers prior to VAT adoption. Tax incidence is represented by firm-specific tax betas, i.e., selling price-tax sensitivities, product cost-tax sensitivities and input cost-tax sensitivities. We first define a firm-level sales tax rate by dividing sales tax a firm paid in a year by the total sales. Then we compute *Beta_Sales_Price* as the standardized beta from regressing sales-value-weighted selling price on firm-level sales tax rate over the pre-VAT period. *Beta_Product_Cost* is the standardized beta from regressing production-value-weighted product cost on firm-level sales tax rate over the pre-VAT period. *Beta_Input_Cost* is the standardized beta from regressing input-value-weighted input cost on firm-level sales tax rate over the pre-VAT period. To interpret higher values as indication of more of the incidence being passed on to suppliers, we multiply *Beta_Product_Cost* and *Beta_Input_Cost*, respectively by -1 . We denote these variables as *Inverse_Beta_Product_Cost* and *Inverse_Beta_Input_Cost*, respectively. Standard errors (reported in parentheses) are clustered at the state-event level. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively.

Dependent Variable	(1) <i>Products</i>	(2) <i>Products</i>	(3) <i>Products</i>	(4) <i>Products</i>	(5) <i>Products</i>	(6) <i>Products</i>
VAT	−0.262*** (0.067)	–	−0.438*** (0.094)	–	−0.401*** (0.081)	–
VAT × <i>Beta_Sales_Price</i>	0.253** (0.125)	0.305** (0.118)				
VAT × <i>Inverse_Beta_Product_Cost</i>			0.310*** (0.092)	0.280*** (0.095)		
VAT × <i>Inverse_Beta_Input_Cost</i>					0.269* (0.134)	0.258* (0.132)
<i>Beta_Sales_Price</i> × Post	0.201*** (0.054)	0.189*** (0.054)				
<i>Inverse_Beta_Product_Cost</i> × Post			−0.139*** (0.052)	−0.135** (0.051)		
<i>Inverse_Beta_Input_Cost</i> × Post					−0.063 (0.052)	−0.056 (0.047)
Controls	Y	Y	Y	Y	Y	Y
Industry × Year FE	Y	Y	Y	Y	Y	Y
Firm FE	Y	Y	Y	Y	Y	Y
State × Year FE	N	Y	N	Y	N	Y
Observations	19,648	19,648	15,592	15,592	18,872	18,872
R-squared	0.877	0.879	0.874	0.876	0.880	0.882

Table 8

Heterogeneity by Product Labor Intensity

This table reports the results of the effect of the VAT on product mix conditional on labor intensity defined based on products. *High_Labor_Intensity* is an indicator for firms with a larger-than-median level of product labor intensity. To construct this measure, we first obtain labor intensity for each single-product firm, defined as employee compensation scaled by total assets. Then, we average across products of a firm to obtain an average product labor intensity. Standard errors (reported in parentheses) are clustered at the state-event level. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively.

	(1) <i>Products</i>	(2) <i>Products</i>
VAT	−0.075 (0.061)	–
VAT × <i>High_Labor_Intensity</i>	−0.364*** (0.134)	−0.363** (0.142)
<i>High_Labor_Intensity</i> × Post	0.116* (0.069)	0.096 (0.077)
Controls	Y	Y
Industry × Year FE	Y	Y
Firm FE	Y	Y
State × Year FE	N	Y
Observations	37,090	37,090
R-squared	0.858	0.858

Table 9

VAT and Corporate Efficiency Outcomes

This table provides evidence on the effect of the VAT on corporate efficiency outcomes, including product manufacturing cost (Panel A), firm performance and firm value (Panel B), and investment efficiency (investment-sensitivity) (Panel C). *Unit Manufacturing Cost* is estimated as the natural logarithm of the total value of ending inventory divided by units of ending inventory for a firm-product in a year. *Q* is Tobin's *Q*, which is defined as the market value of equity plus the book value of debt divided by total assets. Standard errors (reported in parentheses) are clustered at the state-event level. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively.

Panel A. Product cost		
	(1)	(2)
Dependent variable	<i>Unit Manufacturing Cost</i>	<i>Unit Manufacturing Cost</i>
VAT	−0.066*** (0.023)	−0.044** (0.017)
Controls	Y	Y
Industry × Year FE	Y	Y
Firm FE	Y	Y
Product Type FE	N	Y
Observations	79,277	79,277
R-squared	0.512	0.778
Panel B. Firm profitability and firm value		
	(1)	(2)
Dependent variable	ROA	Q
VAT	0.009*** (0.001)	0.303*** (0.062)
Controls	Y	Y
Industry × Year FE	Y	Y
Firm FE	Y	Y
Observations	41,209	41,209
R-squared	0.532	0.632
Panel C. Investment efficiency		
	(1)	(2)
Dependent variable	<i>Investment</i>	<i>Investment</i>
Q	0.033*** (0.005)	0.028*** (0.004)
VAT × Q	0.016** (0.006)	0.010** (0.005)
VAT	−0.038*** (0.011)	–
Controls	Y	Y
Industry × Year FE	Y	Y
Firm FE	Y	Y
State × Year FE	N	Y
Observations	37,938	37,938
R-squared	0.520	0.533

across all products of a firm for the period prior to the tax reform. A higher *Product HHI* indicates that a firm operates in more concentrated markets. We then expand Equation (1) by interacting *VAT* and *Post* with *Product HHI*. Panel A of Table 6 presents the results. Column 1 reports the baseline specification, while column 2 includes state-year fixed effects. We find a negative *VAT* coefficient in column 1, indicating that in less concentrated markets, firms reduce the number of product offerings. Importantly, the interaction term *VAT* × *Product HHI* is negative and significant in both columns, consistent with the prediction that firms in more concentrated product markets respond more strongly to the VAT adoption by narrowing their product scope.

Next, we corroborate our findings on the role of tax incidence by exploring variation *within* product markets. Building on tax incidence theory, the model by Weyl and Fabinger (2013) shows that, regardless of market structure—competitive or concentrated—the relative elasticity of supply and demand determines who bears the tax burden. We therefore predict that firms that bear more (less) of the consumption tax will reduce their product offerings to a greater (smaller) extent. To test this empirically, we examine differences in firms' markups, *Firm Markup*, using markups as a proxy for the relative elasticity of supply and demand. Higher markups

indicate that firms can pass more of the tax burden to customers or suppliers, thus bearing less of the burden themselves.²⁴ Accordingly, we expect a smaller (larger) effect of the VAT adoption on product mix for firms with higher (lower) markups.

We construct our measure of markups as follows. For each product of a firm, we calculate the markup as sales minus cost divided by cost. A firm with multiple products thus has multiple markups. To make these markups comparable across product types, we normalize them by subtracting the minimum product-market-wide markup and dividing by the range between the maximum and minimum product-market-wide markups. For each firm, *Firm Markup* is then calculated as the average of the normalized markups over the pre-reform period. A higher *Firm Markup* indicates a higher average product markup. We use this measure and expand Equation (1) by including its interaction with *VAT* and *Post*.

The results of this test are reported in Panel B of Table 6. Consistent with our predictions, we find a negative and significant coefficient on *VAT* and a positive and significant coefficient on *VAT* $\times$ *Firm Markup* in column 1. Firms that bear more of the tax burden (i.e., those with lower markups) reduce their product scope, as indicated by the negative *VAT* coefficient. For firms that bear less of the consumption tax (i.e., those with higher markups), the reduction in product scope is smaller, as reflected in the positive *VAT* $\times$ *Firm Markup* coefficient. The *VAT* $\times$ *Firm Markup* coefficient remains similar and significant when including state-year fixed effects (column 2). These findings indicate that firms are more likely to cut products when they bear a greater share of the consumption tax burden.

5.1.2. Firm-level measures of tax incidence

In the second set of tests related to tax incidence, we leverage our data on firm-level product prices and sales prices, as well as the reported sales taxes paid by firms prior to the VAT adoption. Using these data, we construct ex ante firm-level measures of tax incidence that capture the degree of pass-through of taxes to customers and suppliers, respectively. Conceptually, a higher sensitivity of sales prices to sales taxes (a lower sensitivity of production costs to sales taxes) implies that less of the sales tax burden is borne by the focal firm's shareholders and that the sales tax burden is partly passed on to consumers (suppliers).

We construct these measures in the following steps. For each firm, we first regress sales prices, weighted by sales values, on effective sales tax rates (i.e., sales tax payments scaled by total sales) for the years preceding the VAT adoption. The resulting firm-specific coefficients on the effective sales tax rate capture the sensitivity of sales prices to taxes. A higher coefficient indicates that firms can pass on more of the sales tax burden to consumers through higher prices. We standardize these coefficients to have a mean of 0 and a standard deviation of 1, denoting this measure as *Beta_Sales_Price*. Similarly, to capture the extent to which firms can shift the sales tax burden to suppliers, we construct *Beta_Product_Cost*. To obtain this measure, we replace sales price with product cost as the dependent variable in the first step. Because a lower coefficient for *Beta_Product_Cost* implies that more of the tax burden is passed on to suppliers via lower product costs, we multiply it by -1 . This measure, denoted *Inverse_Beta_Product_Cost*, can be interpreted analogously to *Beta_Sales_Price*: a higher value of *Inverse_Beta_Product_Cost* indicates that firms bear less of the sales tax burden and pass more on to suppliers. Since product cost aggregates raw materials, labor, and manufacturing overheads, we focus specifically on material costs to

Table 10

Heterogeneity by Managers' Empire Building Incentives

This table reports the results of the effect of the VAT on product mix conditional on the managers' empire-building incentives, proxied by FCF. *FCF* is cash flow from operating activities minus dividends paid scaled by total assets. Standard errors (reported in parentheses) are clustered at the state-event level. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively.

	(1)	(2)
	<i>Products</i>	<i>Products</i>
<i>VAT</i>	−0.303*** (0.076)	-
<i>VAT</i> $\times$ <i>FCF</i>	1.910* (1.058)	1.995* (1.006)
<i>FCF</i> $\times$ <i>Post</i>	0.664* (0.366)	0.490** (0.233)
Controls	Y	Y
Industry $\times$ Year FE	Y	Y
Firm FE	Y	Y
State $\times$ Year FE	N	Y
Observations	40,565	40,565
R-squared	0.856	0.856

²⁴ We note that *Firm Markup* and *HHI* capture different concepts. *Firm Markup* reflects variation in firms' markups *within* a market, whereas *HHI* captures variation in market concentration *across* markets. Consistently, the average *Firm Markup* is 0.8 in both less concentrated and more concentrated markets.

capture supplier exposure more directly. We define *Inverse Beta Input Cost* based on firm-level material costs as an alternative measure of the extent to which suppliers bear part of the firm's sales tax burden on input goods.

We then use these three firm-specific measures and interact each individually with *VAT* and *Post*. The results are reported in Table 7. Across all tests, the coefficients on *VAT* remain negative and statistically significant. Importantly, we find positive and significant coefficients on the interactions of *VAT* with *Beta Sales Price*, *Inverse Beta Product Cost*, and *Inverse Beta Input Cost*. These results hold with and without state-year fixed effects. Overall, these findings corroborate the evidence from Section 5.1.1 and indicate that the effect of VAT adoption on product mix is attenuated when either consumers or suppliers bear a larger share of the sales tax burden on input goods.

5.2. Heterogeneity by labor intensity

Finally, we explore the role of labor intensity. Under a cascading sales tax, the incentive to vertically integrate is stronger for labor-intensive production processes because externally sourced inputs are subject to higher markups, exacerbating tax cascading. Under a VAT regime, labor-intensive firms may be more likely to outsource production, as they can claim VAT input credits on labor-heavy products. We therefore expect a stronger effect of the VAT adoption on product mix for firms with labor-intensive products. To measure labor intensity, we first focus on single-product firms and calculate product-specific labor intensity as the average ratio of employee compensation to total assets. We then apply this product-specific labor intensity measure to all sample firms by averaging across their products to obtain firm-level product labor intensity. We then define *High Labor Intensity* as an indicator, which equals one for firms with an above-median level of product labor intensity. Interacting *High Labor Intensity* with *VAT* and *Post*, we rerun our main model. The results are reported in Table 8. We find that firms with more labor-intensive products respond more strongly to VAT adoption. The result remains robust when including state-year fixed effects to account for unobservable state-year characteristics. This finding is consistent with the importance of labor costs in product pricing: under the sales tax, labor costs increase tax cascading and incentivize vertical integration, whereas under VAT, input tax credits reduce this incentive.

6. Supplemental tests

6.1. VAT and corporate efficiency outcomes

We extend our analysis beyond product mix decisions to examine the effect of VAT adoption on overall corporate outcomes. As discussed in Section 2, firms maximize value under a sales tax by vertically integrating and producing input goods to avoid cascading taxes. However, such tax-induced vertical integration may be operationally inefficient, for example, due to reduced specialization (Williamson, 1971; Grossman and Hart, 1986). In other words, sales tax-driven vertical integration may reflect an inefficient allocation of resources across firms. Instead of specializing in their most profitable products, firms may vertically integrate due to the cascading sales tax. The transition to VAT alters the cost–benefit trade-offs of vertical integration by removing cascading tax distortions. Under VAT, firms can implement a level of vertical integration closer to the non-tax optimal, thereby aligning their operations with a more efficient pre-tax product mix. We investigate whether the observed responses of the product mix to VAT adoption lead to higher profitability and, ultimately, increased firm value in three steps.

First, we examine the effect of VAT adoption on manufacturing costs. If the sales tax inefficiently locks production capacity into multi-segment firms, the VAT adoption should reduce manufacturing costs as external suppliers can produce the input goods at lower costs. To this end, we analyze manufacturing costs at the product level, capturing all costs involved in converting inputs into finished goods (e.g., for machinery, financing, and wages). Specifically, we define the manufacturing cost per unit as the unit cost of ending inventory. Following this approach, we calculate *Unit Manufacturing Cost* for each firm-product-year as the natural logarithm of the total value of ending inventory divided by the number of units in ending inventory. We then use *Unit Manufacturing Costs* as the dependent variable in our baseline specification. The results (column 1, Panel A, Table 9) indicate that VAT adoption reduces manufacturing costs by over 6.6%. In column 2, we further control for unobserved product-specific characteristics likely to influence unit costs by including product category fixed effects. The coefficient on *VAT* remains similar in magnitude and statistically significant with these high-dimensional fixed effects. Collectively, these results suggest that VAT adoption lowers manufacturing costs, consistent with the notion that cascading sales taxes previously incentivized firms to produce input goods internally, even when less efficient.

Next, we examine the effect of VAT adoption on overall firm profitability and value. We measure profitability using *ROA* and firm value using Tobin's *Q* (*Q*). The results (Panel B, Table 9) show that VAT adoption increases profitability and firm value. These findings are consistent with the sales tax previously inducing inefficient vertical integration. By eliminating cascading tax distortions, firms operate more efficiently, which is reflected in higher profitability and greater firm value.

Finally, we examine investment efficiency. Under a cascading sales tax, firms face constraints in responding to new investment opportunities because expanding requires investments not only in the final good but also in input goods, making vertical expansion costly. In contrast, under a VAT regime, firms can more easily respond to opportunities, focusing on key products while flexibly outsourcing or producing inputs internally. Following Whited (1992), Baker et al. (2003), and Mclean, Zhang, and Zhao (2012), we test whether investment-*Q* sensitivity strengthens after VAT adoption. We use capital expenditure (*Investment*) as the dependent variable, with *Q* and the interaction of $VAT \times Q$ as the main independent variables of interest. The results (Table 9, Panel C) show that VAT adoption improves investment efficiency. We find a positive coefficient on *Q* suggesting that firms respond to investment opportunities. Importantly, we find a positive coefficient on the interaction term of $VAT \times Q$, suggesting that investment-*Q* sensitivity (i.e., investment efficiency) increases post-VAT. With fewer frictions in the consumption tax system, firms become more responsive to

investment opportunities. The economic magnitude of the estimates indicates that the investment-Q sensitivity increases by 48% ($=0.016/0.033$) after VAT adoption. Importantly, when controlling for state-year fixed effects, the coefficient on $VAT \times Q$ remains significant and very similar.

Taken together, these findings suggest that the sales tax induced an inefficient level of vertical integration, forcing firms to internalize production to avoid cascading effects. After VAT adoption, firms can vertically disintegrate and determine their scope based on nontax considerations. This shift reduces production costs, increases profitability and firm value, and enhances investment efficiency.

6.2. The role of agency issues

While our findings of reduced product scope following VAT adoption are consistent with efficient vertical disintegration from the shareholders' perspective, not all firms may respond this way due to agency issues between owners and managers (Jensen, 1986). That is, managers with empire-building incentives may resist vertical disintegration after VAT adoption. Accordingly, we expect an attenuated effect of VAT on product mix in firms where such managerial incentives are stronger. We proxy for managerial empire-building using free cash flow, *FCF*, defined as cash flow from operating activities minus dividends paid, scaled by total assets. Higher *FCF* indicates a greater likelihood of managerial empire-building (Chen et al., 2012). Table 10 presents the results, where we rerun Equation (1) and interact *VAT* and *Post* with *FCF*. The positive and significant coefficients on $VAT \times FCF$ support the conjecture that the effect of VAT on product mix diminishes with higher managerial empire-building incentives. These findings remain robust after including state-year fixed effects to account for potential state-level confounders.

7. Conclusion

This paper exploits unique firm-level data with detailed product-level information to examine how consumption tax systems—in particular, tax burdens on input goods—affect firms' product mix decisions. We exploit a shift from a cascading sales tax to a VAT system in India. We find that the VAT adoption, which substantially reduced the tax burden on input goods, led firms to narrow their product category scopes by facilitating vertical disintegration. We corroborate this finding by constructing a novel firm-level vertical integration measure, leveraging detailed product descriptions and LLMs. We also show that firms concentrate on their best-performing products post-VAT, indicating that the reform encourages product specialization. These strategic adjustments—product scope narrowing and specialization—enhance overall firm efficiency, as reflected in lower manufacturing costs, higher profitability, greater firm value, and improved investment efficiency under the VAT system relative to the previous cascading sales tax regime. Cross-sectional analysis further reveals that the product mix response is attenuated when the burden of the former sales tax falls more on consumers or suppliers than on the focal firms.

These findings carry important policy implications. We show that a consumption tax regime that exempts input goods from taxation promotes a more efficient allocation of resources across firms. By contrast, a system that taxes inputs can create inefficient “lock-in” effects, incentivizing firms to maintain vertically integrated, multi-product structures solely to avoid the cascading burden of multiple sales taxes. A well-functioning VAT system, which provides timely and full input tax credits, can eliminate these distortions. This is particularly crucial for developing economies, where administrative challenges often delay the provision of such credits (IMF, 2021). Our results highlight that improving the consumption tax administration—especially by ensuring prompt and full input tax credits—can significantly enhance inter-firm resource allocation.

While our findings highlight the role of consumption tax design in shaping firms' production choices, our approach comes with limitations. Our empirical setting—India's staggered VAT adoption—offers high internal validity, and our results stem from a single-country context and a reform that took place in the early 2000s. Moreover, the reform represented a substantial shift in the tax system, which may have triggered a stronger business response than a more incremental policy change. Nonetheless, the mechanism we document—distortions arising from a tax burden on input goods—is unlikely to be unique to India. Because tax burdens on input goods can also arise under existing sales tax or VAT systems (IMF, 2021), the economic mechanism we uncover may be relevant in other contexts. Future research should explore the extent to which similar dynamics occur in different institutional environments, such as developed versus developing economies.

Declaration of competing interest

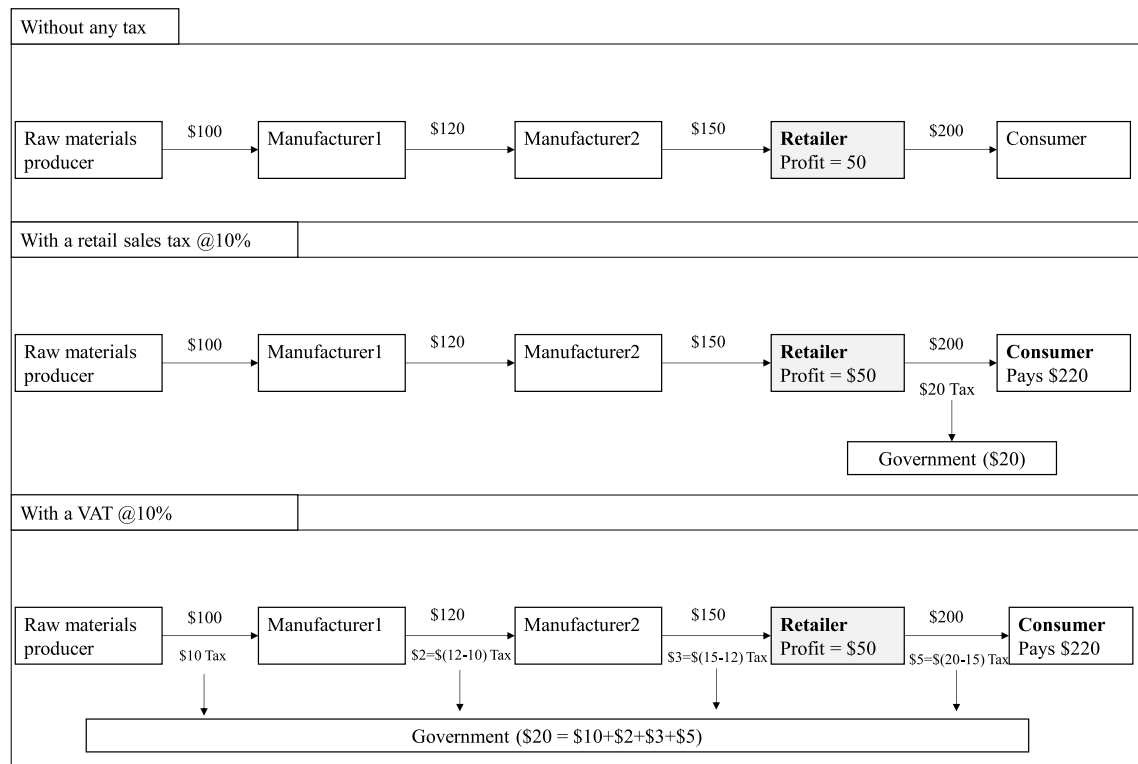
We have nothing to disclose.

Appendix C. Supplementary data

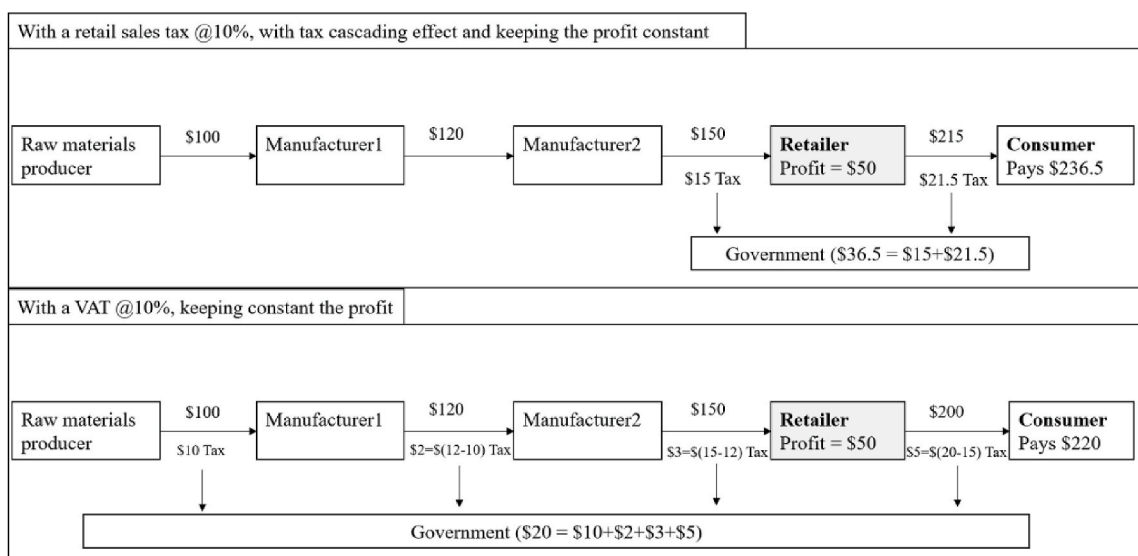
Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jacceco.2026.101898>.

Appendix A An Illustrative Example on Tax Cascading Effect under Sales Tax and the Elimination of Cascading Effect under VAT

Panel A: Ideal Sales Tax and VAT



Panel B: Extended Example with Cascading Sales Tax



This figure illustrates the tax cascading effect under a sales tax, and how the VAT adoption eliminates the tax cascading effect and the subsequent input price distortions. The focal firm is the retailer. In Panel A, we illustrate in the ideal world what it means for tax payments and manufacturing cost when we shift from a non-tax regime to a sales tax regime versus a VAT regime. In the ideal world, consumption tax, regardless of it being a sales tax or a VAT, is paid effectively only once. In Panel A, end consumers pay effectively a 10% tax rate (equivalent to \$20 for a product sold at \$200 to consumers) under both the sales tax and the VAT. In both cases, the firm earns a profit of \$50 and the consumers pay \$220 for the product.

In Panel B, we illustrate the case where sales tax may be charged at several steps of the production process (not just the last step), which results in a tax cascading effect. This is partly driven by the difficulty in distinguishing between sales for production purpose or consumption purposes (Hoseini and Briand, 2020). Suppose a nominal sales tax rate and VAT are both set at 10%. For a bounded solution, we assume the profit of imposing a sales tax and VAT is the same for each production stage participant.

In the refined example above, with a sales tax in Panel B,

- Raw material producer sells its product at a price of \$100.
- Manufacturer 1 does not pay sales tax, sells its product at a price of \$120, has a cost of \$100 (from Raw material producer), and ends up with a profit of \$20 ($=\$120-100$)
- Manufacturer 2 pays \$15 ($=\$150*10\%$) sales tax, sells its product at a price of \$165 ($=\$150*(1 + 10\%)$), has a cost of \$120 (from Manufacturer 1), and ends up with a profit of \$30 ($=\$165-120-15$)
- Retailer now has a cost of \$165 = \$150 + \$15 in tax (from Manufacturer 2). In order to continue to realize a profit of \$50, it needs to sell the product at a retail price of \$236.5 ($=\$215*(1 + 10\%)$).

In the refined example above, with a VAT in Panel B,

- Raw material producer pays \$10 ($=\$100*10\%$) VAT, sells its product at a price of \$110 ($=\$100*(1 + 10\%)$)
- Manufacturer 1 pays \$2 ($=\$120-100*10\%$) VAT, sells its product at a price of \$132 ($=\$120*(1 + 10\%)$),²⁵ has a cost of \$110 (from Raw material producer), and ends up with a profit of \$20 ($=\$132-110-2$)
- Manufacturer 2 pays \$3 ($=\$150-120*10\%$) VAT, sells its product at a price of \$165 ($=\$150*(1 + 10\%)$),²⁶ has a cost of \$132 (from Manufacturer 1), and ends up with a profit of \$30 ($=\$165-132-3$)
- Retailer pays \$5 ($=\$200-150*10\%$) in VAT and sells its product at a retail price of \$220 ($=\$200*(1 + 10\%)$) and has a cost of \$150 (from Manufacturer 2). This results in a profit of \$50 ($=\$220-150-5$).

Hence, in order to for Retailer to earn the same profit, a Retailer subject to the cascading sales tax needs to charge a higher retail price than a Retailer subject to the VAT.

Appendix B. Illustration of the Importance of the Tax Burden on Upstream Suppliers

To illustrate the greater importance of the upstream suppliers' sales burden vis-à-vis the consumer's sales tax burden, we consider four cases building on Appendix A. The supply chain is very simple in this case. The firm can either use an external input at a cost of 100 (no vertical integration) or produce this product internally (vertical integration). The firm adds a markup of 100 to the input good, which it will charge the customer. If the firm sources the input externally, it has to pay a sales tax of 10%.

The sales tax is imposed on the customer during the point of sale. In the first two cases, we assume that the end-consumer also resides in the same state. We assume a sales tax rate of 10%. In the third and fourth cases, we assume that the end-consumer resides in a state with a sales tax rate of 5%.

Figure A.1 shows the tax burden when the firm sources the input good externally and sells it at home at a price of 210 (100 in input costs plus 10 in sales tax plus 100 in markup). In this simplified example, the consumer now has to pay 231 because she has to pay 21 in sales tax on a value of 210.

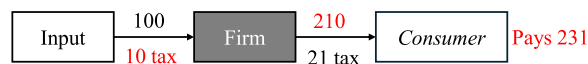


Fig. A.1. Tax cascading on inputs, sale at home.

Figure A.2 illustrates the advantage of vertical integration. Assuming the same cost of 100 for internal production, firms can avoid the cascading sales taxes and charge 200 plus 10% sales tax to the customer. That is, the customer pays 220, and the advantage of vertical integration is 11 in this example.

²⁵ We obtain \$120 by solving $n*(1 + 10\%)-110-(n*10\%-10) = \20 to obtain the same profit as under the sales tax.

²⁶ We obtain \$150 by solving $p*(1 + 10\%)-132-(p*10\%-10-2) = \30 to obtain the same profit as under the sales tax.

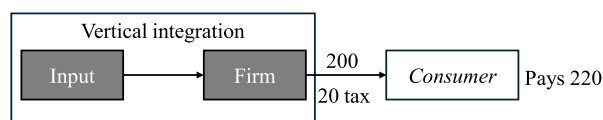


Fig. A.2. No tax cascading (vertical integration), sale at home.

In Figures A.3 and A.4, these cases are replicated for out-of-state sales. In the case of external inputs, the consumer will pay 220.5, as the tax base of 210 is now only multiplied by 5%. In the case of vertical integration, the firm can reduce the price for the consumer to 210. That is, vertical integration can give the firm a considerable advantage. Importantly, the advantage of vertical integration is 10.5 for an out-of-state sale, which is very similar to the advantage of 11 for a within-state sale. When moving to a VAT, firms have an incentive to vertically disintegrate because they will receive a tax credit on external inputs of 10 irrespective of whether they ultimately make a within-state or out-of-state sale.

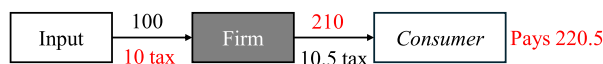


Fig. A.3. Tax cascading on inputs, sale in another state.

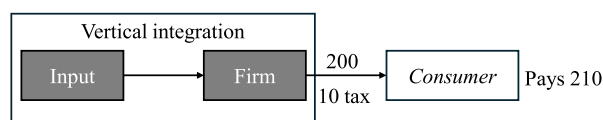


Fig. A.4. No tax cascading (vertical integration), sale in another state.

Hence, the advantage of vertical integration—or the cost of outsourcing the production of input and intermediary goods—is mostly determined by the sales tax on the upstream input. Because the latter is determined by the headquarters' location, the headquarters' location and not the customer's location is decisive for the firm's decision to vertically integrate.

Appendix Table A1
Variable definition

Variable	Definition
<i>Products</i>	Number of unique products a firm produces in a year
<i>VAT</i>	An indicator variable that equals one for the year in which an Indian state adopted a VAT and for all subsequent years, and zero otherwise. VAT is effectively the interaction of <i>VAT State</i> and <i>Post</i>
<i>VAT State</i>	An indicator variable that equals one for the Indian state that adopted a VAT for the respective event, and zero otherwise
<i>Post</i>	An indicator variable that equals one for post VAT adoption year for the respective event, and zero otherwise
<i>Proportion of Continued Products</i>	Number of unique products that are continued to be manufactured in the next n (1, 2, 3, ...) years, divided by the total number of unique products that are manufactured at the beginning of each event window
<i>Firm_VI</i>	Firm-year vertical integration. We calculate firm-level integration following the three steps. First, we use GPT-4o-mini to identify the input–output relationship between any two products for the universe of products in the database, which is illustrated in Online Appendix, Part B. We end up with product pairwise data. For every product A in the product pair, we identify product B as either upstream or downstream or irrelevant. Second, we identify the five main products of any given firm, defined based on the firm-product level sales. Third, for each one of the main products of a firm, we obtain the extent to which other non-main products are vertically related to the main product by calculating the ratio of vertically related products to total number of non-main products. <i>Firm_VI</i> is then calculated as the average value of the ratios with respect to those main products
<i>Firm_VI Weighted</i>	Weighted measure of <i>Firm_VI</i> , with weights (in %) being the proportion of sales volume of each of the products relative to total sales
<i>Within-Type Products</i>	The number of subtypes of products for a given firm-product type in a year
<i>Product Concentration</i>	The sum of all squared portfolio shares of different products a firm produces in a year. A higher (smaller) value means that products produced are more concentrated (diversified). Product portfolio share is calculated as the ratio of revenue generated from each product of a firm to the total revenues generated from all products of the firm
<i>Markup-Weighted Product Concentration</i>	The sum of weighted squared portfolio shares of different products a firm produces in a year, with weights being the markup of each product in a product portfolio. Markup is defined as sales minus cost divided by cost for each product
<i>No. of Subsidiaries</i>	The log of the number of subsidiaries belonging to a sample firm
<i>Unit Manufacturing Cost</i>	The natural logarithm of the total value of ending inventory divided by units of ending inventory for a firm-product in a year
<i>Outsourcing</i>	The ratio of the cost of purchased goods to the sum of the cost of purchased goods and the manufacturing costs
<i>Log_Products</i>	The natural logarithm of number of products produced by a firm in a year
<i>Log_asset</i>	The natural logarithm of total assets
<i>Leverage</i>	Total book value of debt divided by the book value of assets
<i>Sales_growth</i>	Change in sales divided by sales in the last period
<i>ROA</i>	Operating profit divided by total assets
<i>Q</i>	Tobin's Q, which is defined as the market value of equity plus the book value of debt divided by total assets

(continued on next page)

Appendix Table A1 (continued)

Variable	Definition
<i>MB</i>	Market value of equity divided by the book value of equity
<i>RD</i>	R&D expense divided by total assets. The value is replaced with 0 if the data are missing
<i>MissingRD</i>	An indicator variable equal to one if R&D expense information is missing
<i>Investment</i>	Capital expenditures divided by total assets
<i>SD_ROA</i>	Standard deviation of ROAs in a five-year rolling window
<i>SEZ</i>	An indicator for state-years with newly established special economic zone projects
<i>NREGA</i>	An indicator for state-years with National Rural Employment Guarantee Act
<i>FDI</i>	The log of foreign direct investment into states over the year
<i>Opp_party</i>	An indicator for states which are ruled by opposition parties to the federal government over the year
<i>GQ</i>	An indicator for firms with entry into Golden Quadrilateral highway for a distance of less than 50 km after 2006 (Naaraayanan and Wolfenzon, 2024)
<i>GQ_dist</i>	The log of distance to Golden Quadrilateral highway system, which is a continuous variable defined based on GQ
<i>State_econ_inc</i>	An indicator for state-years with economic and tax incentives
<i>Natural_disaster</i>	The log of deaths from natural disasters for state-years

Appendix Table A2

Estimating Elasticity of Product Mix to Hypothetical Cascading Taxes

This table presents the results of estimating the elasticity of product mix with respect to cascading tax rates. *Log_Products* is the log of *Products*. *Products* is the number of unique products a firm produces in a year. *VAT Exposure* is constructed as a measure that captures the hypothetical cascading tax burden vertically-integrated firms would have faced in the absence of integration prior to the VAT reform. We calculate the hypothetical cascading taxes faced by vertically-integrated firms in several steps. First, we obtain product-level sales tax rates on inputs of horizontally-integrated, single-product firms in the pre-VAT reform period. These rates are obtained by dividing sales tax paid by total sales (total assets) in columns 1 and 2 (columns 3 and 4). We then apply these rates to the manufacturing costs of each intermediate input good that is produced within vertically-integrated firms. This approximates the tax cascading costs those firms would have incurred if they were not integrated. The estimated product-level costs are then summed for each firm and scaled by total sales (columns 1 and 2) or by total assets (cols 3 and 4) to create a hypothetical tax rate, which we denote *VAT Exposure*. *VAT Exposure* is always zero for horizontally-integrated firms, as those firms by definition do not have middle products. Standard errors (reported in parentheses) are clustered at the state-event level. ***, **, and * denote significance at the 1%, 5%, and 10%, respectively.

	(1)	(2)	(3)	(4)
	Scaling by total sales		Scaling by total assets	
Dependent variable	<i>Log_Products</i>	<i>Log_Products</i>	<i>Log_Products</i>	<i>Log_Products</i>
<i>VAT Exposure</i>	−1.027*** (0.130)	−0.975*** (0.147)	−0.845** (0.394)	−0.822** (0.370)
Controls	Y	Y	Y	Y
Industry × Year FE	Y	Y	Y	Y
Firm FE	Y	Y	Y	Y
State × Year FE	N	Y	N	Y
Observations	35,884	35,884	35,892	35,892
R-squared	0.822	0.823	0.826	0.827

Appendix Table A3

Robustness Tests

This table presents the results of robustness tests on the main effect of the VAT adoption on product mix. In column 1, *Proportion of Continued Products* is the number of unique products that are continued to be manufactured in the next n (1, 2, 3, ...) years, divided by the total number of unique products that are manufactured at the beginning of each event window. In column 2, we restrict firms to those with single-state operation. In column 3, we use the log of *Products* as the alternative measure. In column 4, we use Poisson estimation as an alternative method. In column 5, we examine the differential effects of VAT on product mix for early adopters and late adopters. In column 6, we drop the states with VAT being lower than sales tax at the nominal rates. Standard errors (reported in parentheses) are clustered at the state-event level. ***, **, and * denote significance at the 1%, 5%, and 10%, respectively.

	(1)	(2)	(3)	(4)	(5)	(6)
	Alternative product mix measure	Single-state operation	Log of Products	Poisson Estimation	Early vs. late adopters	Dropping low-VAT-rate states
Dep. Var.	<i>Proportion of Continued Products</i>	<i>Products</i>	<i>Log_Products</i>	<i>Products</i>	<i>Products</i>	<i>Products</i>
<i>VAT</i>	−0.056** (0.026)	−0.161*** (0.058)	−0.029** (0.012)	−0.029*** (0.010)		−0.158*** (0.041)
<i>VAT × Early</i>					−0.232*** (0.063)	
<i>VAT × Late</i>					−0.143*** (0.053)	
Controls & FE	Y	Y	Y	Y	Y	Y
Observations	41,214	34,801	41,214	41,214	41,214	39,111
R ² /Pseudo R ²	0.506	0.848	0.828	0.404	0.856	0.854

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